

Community Institute of Commerce & Management Studies

(Affiliated to Dr. Manmohan Singh Bengaluru City University, Approved by AICTE, New Delhi
& Recognized by Government of Karnataka)



CERTIFICATE

NAME :

CLASS :

REGISTER NUMBER :

*This is certified to be the bonafied work of the Student in the Mobile
Application Development Laboratory during the academic year 2025-2026.*

*Number of Practicals Certified out of.....in
the subject of*

.....

Teacher In-Charge

.....

Examiner's Signature

.....

Head Of BCA Department

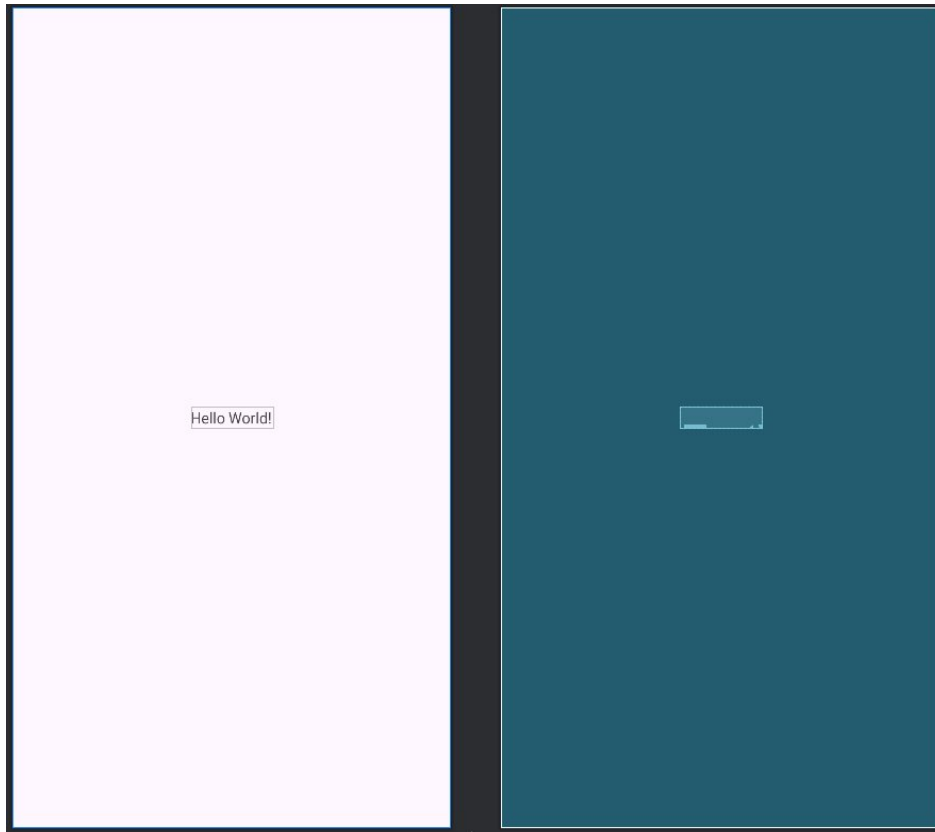
Date :

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Program 01: Creating “Hello World” Application.

Design:



Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textview"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20dp"
        android:text=" Hello World " />

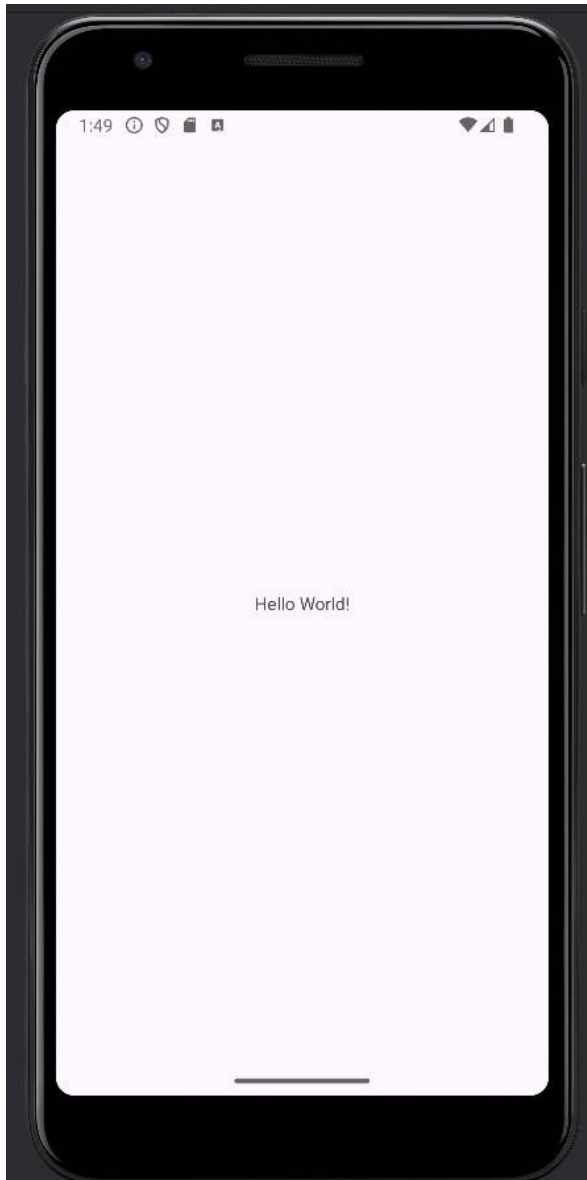
</RelativeLayout>
```

MainActivity.java

```
package com.helloworld;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity1 extends AppCompatActivity
{
    @Override
    protected void onCreate(Bundle savedInstanceState)
    { super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main2);
    }
}
```

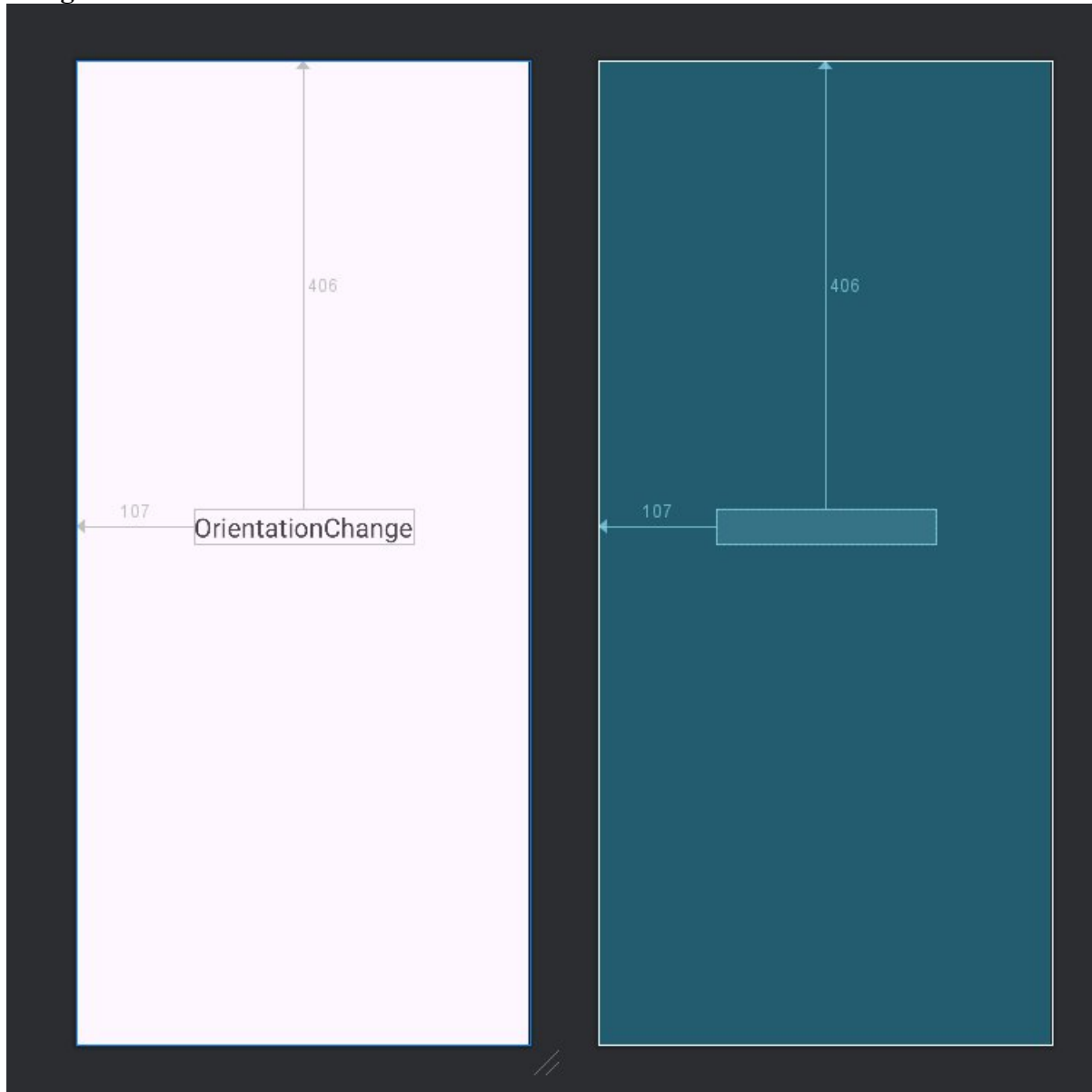
AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/Theme.Helloworld"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>
```

Output:

Program 02: Creating an application that displays message based on the screen orientation

Design:



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
```

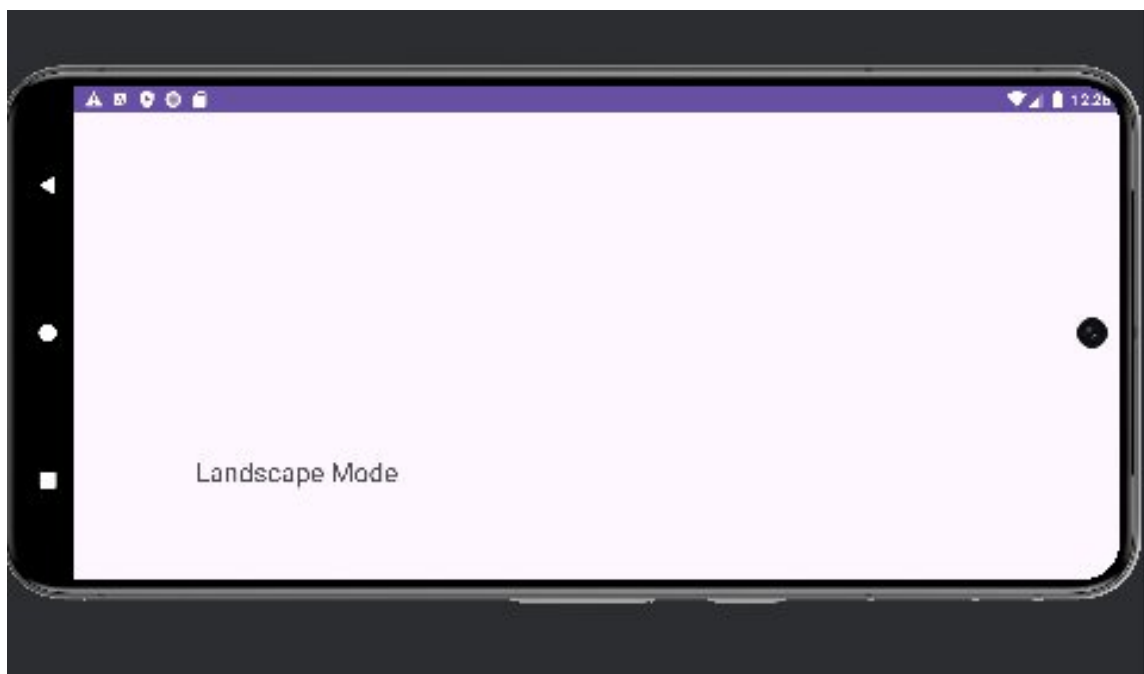
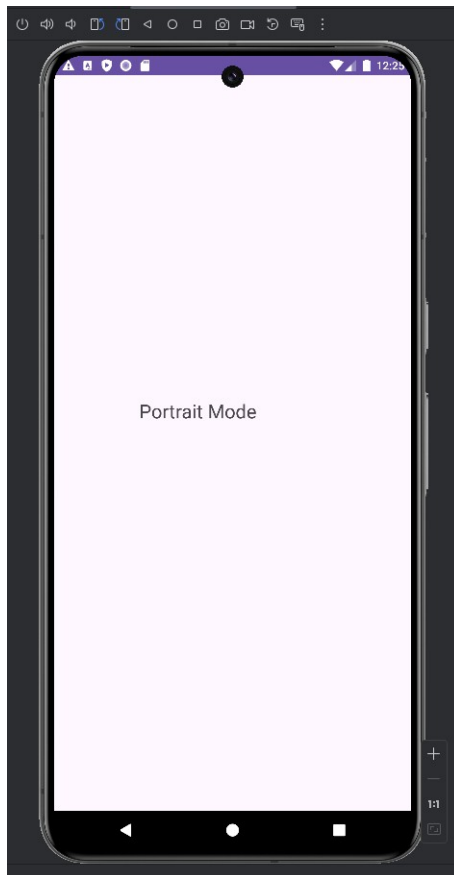
```
<TextView
    android:id="@+id/orientationchange"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentStart="true"
    android:layout_alignParentTop="true"
    android:layout_marginStart="107dp"
    android:layout_marginTop="406dp"
    android:text="OrientationChange"
    android:textSize="24sp" />
</RelativeLayout>
```

MainActivity.java

```
package com.example.orientationchange;
import android.graphics.Point;
import android.os.Bundle;
import android.view.Display;
import android.view.WindowManager;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

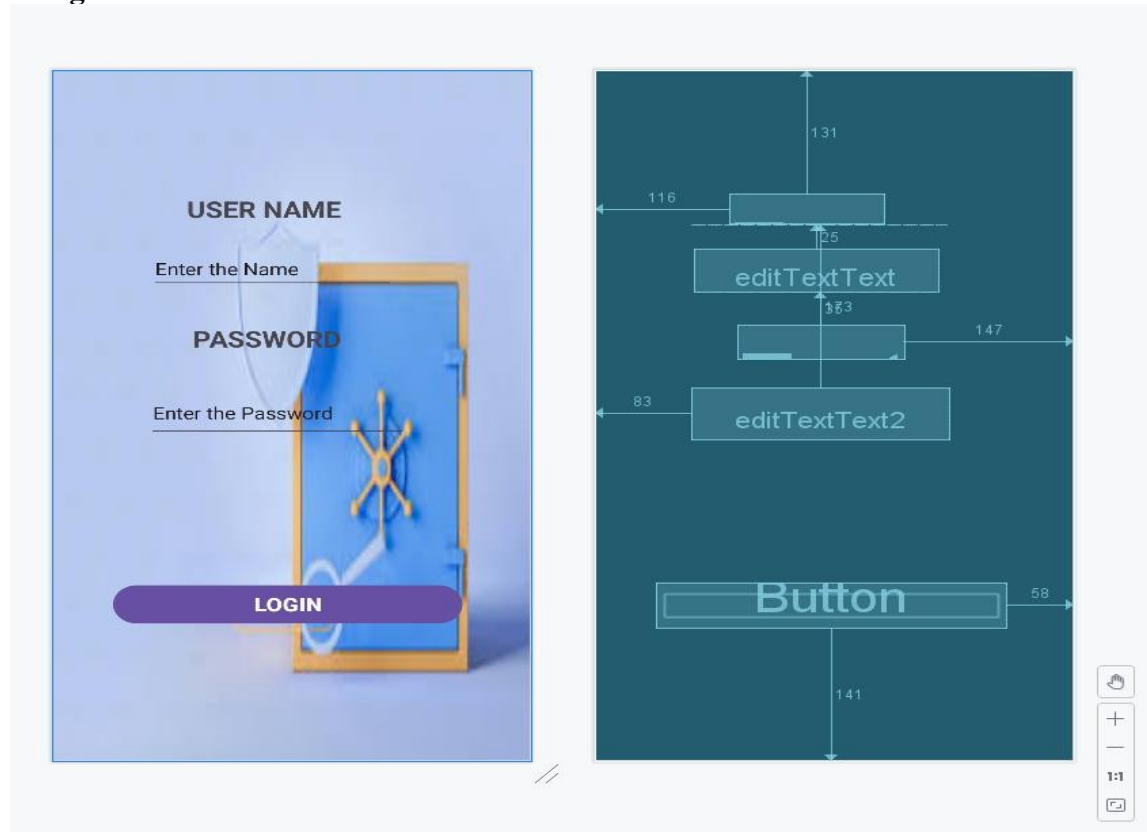
public class MainActivity extends AppCompatActivity
{ @Override
    protected void onCreate(Bundle savedInstanceState)
    { super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      TextView orientationMessage=findViewById(R.id.orientationchange);
      WindowManager wm=(WindowManager) getSystemService(WINDOW_SERVICE);
      Display d=wm.getDefaultDisplay();
      Point size=new Point();
      d.getSize(size);
      int w=size.x;
      int h=size.y;
      if (w>h) {
          orientationMessage.setText("Landscape Mode");
      }
      else {
          orientationMessage.setText("Portrait Mode");
      }
    }
}
```

OUTPUT



Program 03: Developing a Login Window Using UI Controls

Design:



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/images"
    tools:context=".MainActivity">
```

```
<TextView
    android:id="@+id/textView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentStart="true"
    android:layout_alignParentTop="true"
    android:layout_marginStart="116dp"
    android:layout_marginTop="131dp"
    android:text="USER NAME"
    android:textSize="24sp"
    android:textStyle="bold" />
```

```
<EditText
    android:id="@+id/editTextText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/textView"
    android:layout_marginStart="85dp"
```

```

        android:layout_marginTop="25dp"
        android:ems="10"
        android:inputType="text"
        android:text="Enter the Name" />

<TextView
    android:id="@+id/textView2"
    android:layout_width="143dp"
    android:layout_height="36dp"
    android:layout_below="@id/editTextText"
    android:layout_alignParentEnd="true"
    android:layout_marginTop="35dp"
    android:layout_marginEnd="147dp"
    android:text="PASSWORD"
    android:textSize="24sp"
    android:textStyle="bold" />

<EditText
    android:id="@+id/editTextText2"

    android:layout_width="222dp"
    android:layout_height="55dp"
    android:layout_below="@id/textView"
    android:layout_alignParentStart="true"
    android:layout_marginStart="83dp"
    android:layout_marginTop="173dp"
    android:ems="10"
    android:inputType="text"
    android:text="Enter the Password" />

<Button
    android:id="@+id/button"
    android:layout_width="301dp"
    android:layout_height="wrap_content"
    android:layout_alignParentEnd="true"
    android:layout_alignParentBottom="true"
    android:layout_marginEnd="58dp"
    android:layout_marginBottom="141dp"
    android:text="LOGIN"
    android:textSize="20sp"
    android:textStyle="bold" />
</RelativeLayout>

```

MainActivity.java

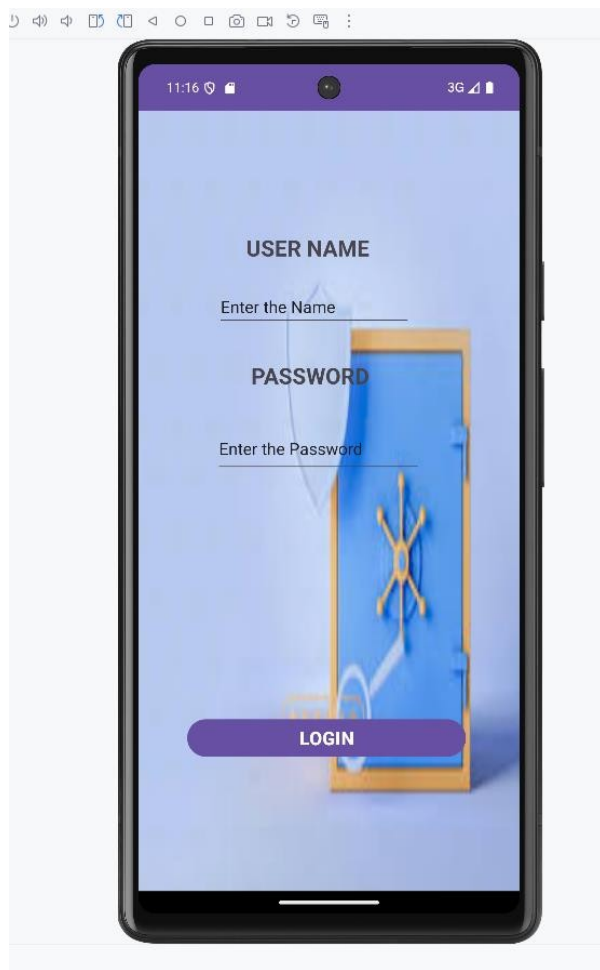
```

package com.example.loginapp;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity
{
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}

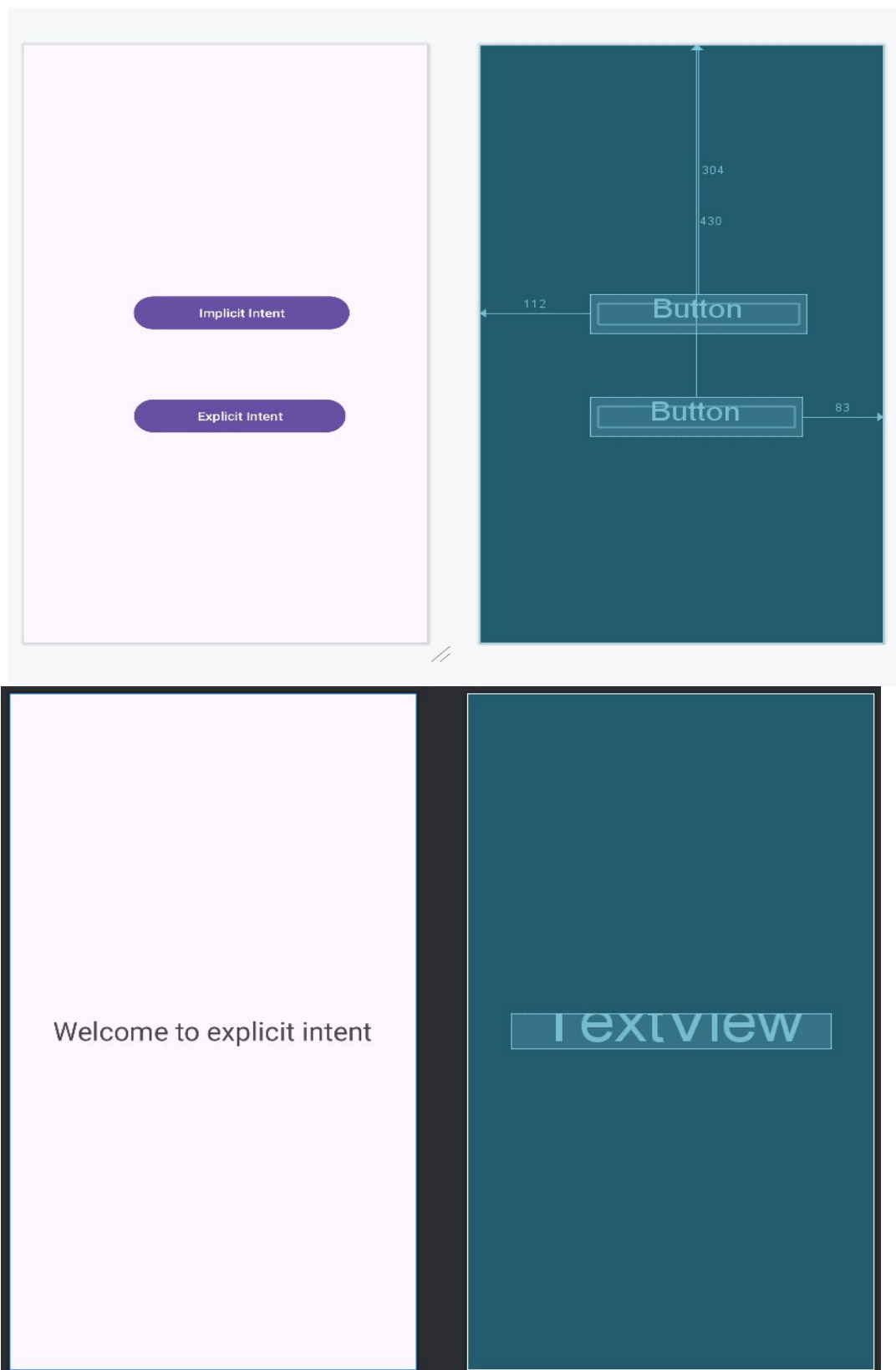
```

OUTPUT



Program 04: Create an application to implement new activity using explicit intent and implicit intent.

Design



Code:**activity_main.xml**

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/button"
        android:layout_width="220dp"
        android:layout_height="wrap_content"
        android:layout_alignParentStart="true"
        android:layout_alignParentTop="true"
        android:layout_marginStart="112dp"
        android:layout_marginTop="304dp"
        android:onClick="onImplicitButtonClicked"
        android:text="Implicit Intent"
        tools:layout_editor_absoluteX="36dp"
        tools:layout_editor_absoluteY="286dp" />

    <Button
        android:id="@+id/button2"
        android:layout_width="216dp"
        android:layout_height="wrap_content"
        android:layout_alignParentTop="true"
        android:layout_alignParentEnd="true"
        android:layout_marginTop="430dp"
        android:layout_marginEnd="83dp"
        android:onClick="onExplicitButtonClicked"
        android:text="Explicit Intent"
        tools:layout_editor_absoluteX="248dp"
        tools:layout_editor_absoluteY="286dp" />

</RelativeLayout>

```

activity_new.xml

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity2">
    <TextView
        android:id="@+id/tv"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textSize="20dp"
        android:text="Welcome to explicit intent"/>

</RelativeLayout>

```

MainActivity.java

```
package com.example.l4;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity
{
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
    public void onImplicitButtonClicked(View view)
    {
        Uri url = Uri.parse("https://www.google.com");
        Intent i = new Intent(Intent.ACTION_VIEW, url);
        startActivity(i);
    }
    public void onExplicitButtonClicked(View view) {
        Intent i=new Intent(MainActivity.this,MainActivity2.class);
        startActivity(i);
    }
}
```

NewActivity.java

```
package com.example.l4;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity2 extends AppCompatActivity
{
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main2);
    }
}
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

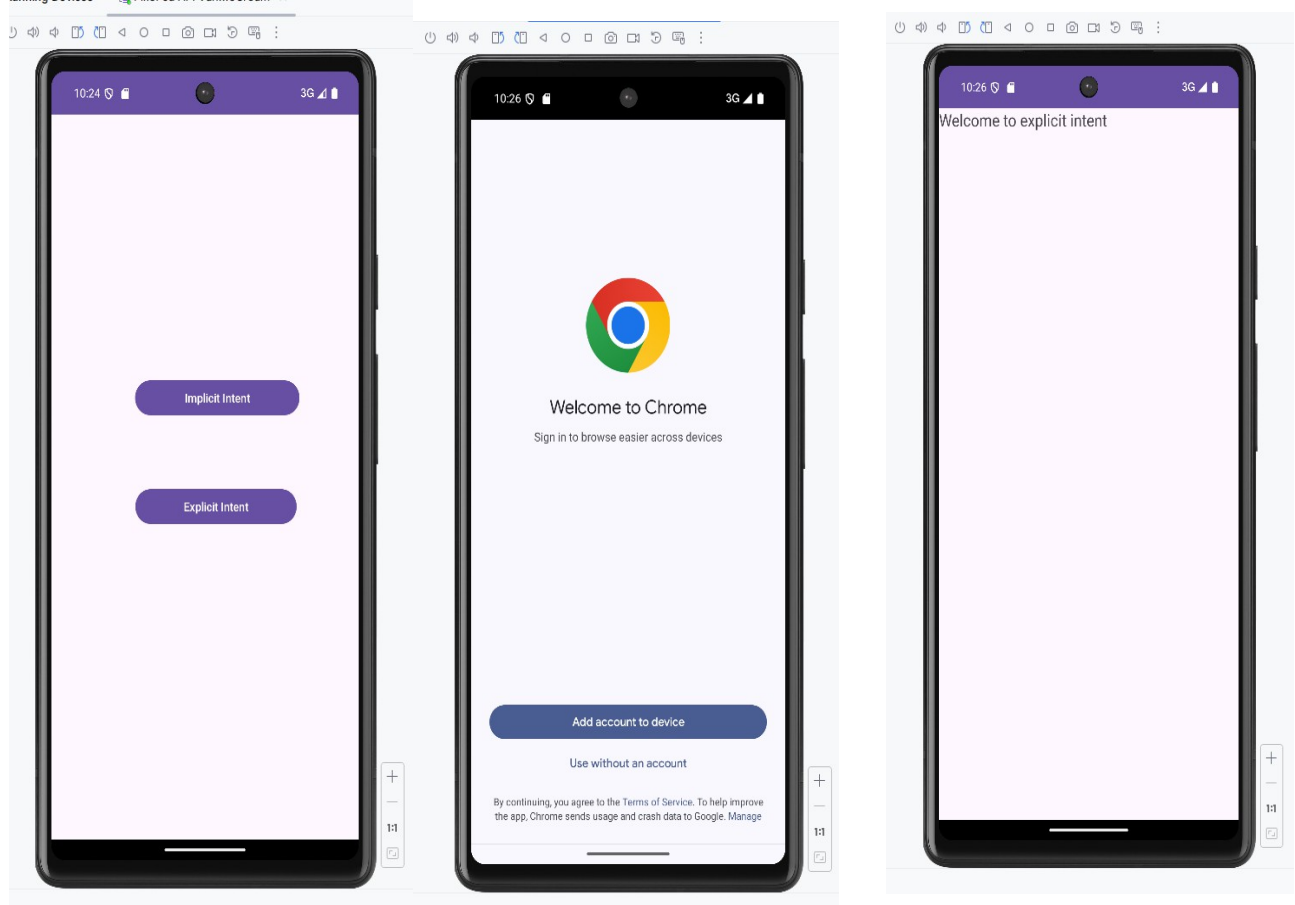
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/Theme.L4"
        tools:targetApi="31">
```

```

<activity
    android:name=".MainActivity2"
    android:exported="false" />
<activity
    android:name=".MainActivity"
    android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />

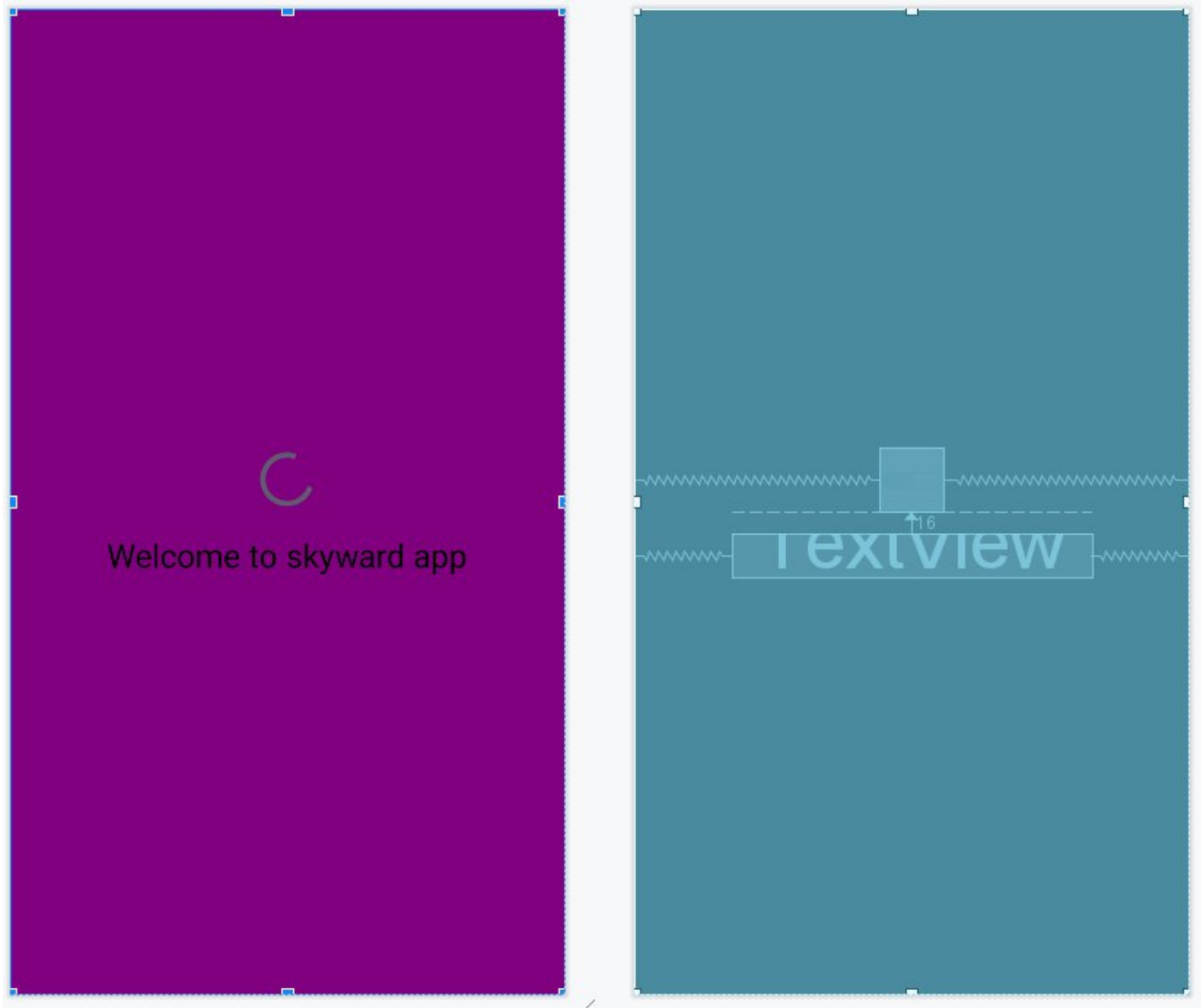
        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>
</application>
</manifest>

```



Program 05: Create an application that displays custom designed Opening Screen

Design



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <RelativeLayout
        android:id="@+id/splash_screen"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:background="#800080"
        android:gravity="center"
        android:padding="16dp">

        <ProgressBar
            android:id="@+id/progress_Bar"
            style="?android:attr/progressBarStyle"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="16dp"
            android:layout_marginTop="16dp"
            android:layout_centerHorizontal="true"
```



```

        android:indeterminate="true"/>

<TextView
    android:id="@+id/textView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Welcome to skyward app"
    android:textSize="24dp"
    android:textColor="#FF000000"
    android:layout_below="@+id/progress_Bar"
    android:layout_centerHorizontal="true"
    android:layout_marginTop="16dp"/>
</RelativeLayout>

<LinearLayout
    android:id="@+id/main_content"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp"
    android:gravity="center"
    android:visibility="gone">

    <TextView
        android:id="@+id/welcome_text"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="welcome to the main activity"
        android:textSize="18sp"
        android:textColor="#0000FF"/>
</LinearLayout>
</FrameLayout>

```

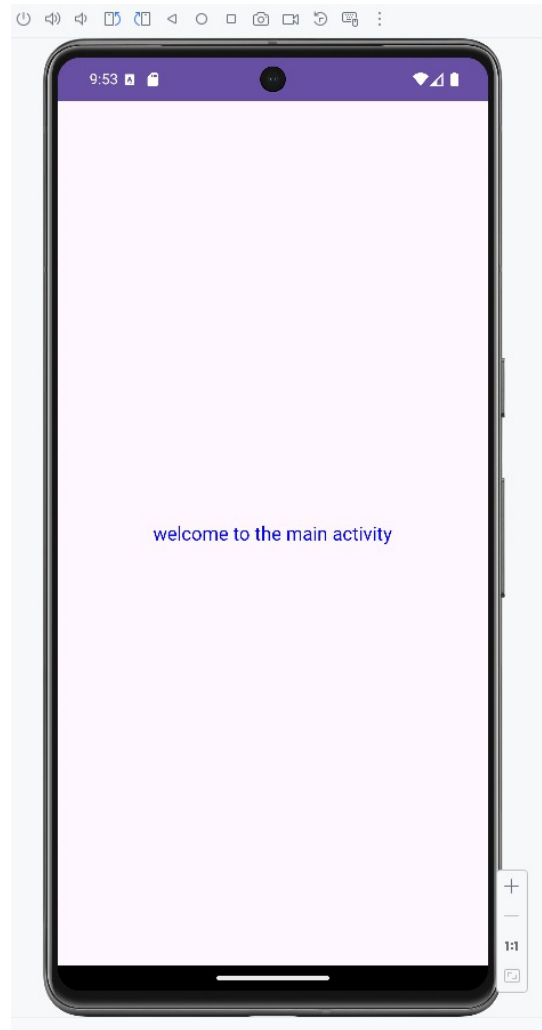
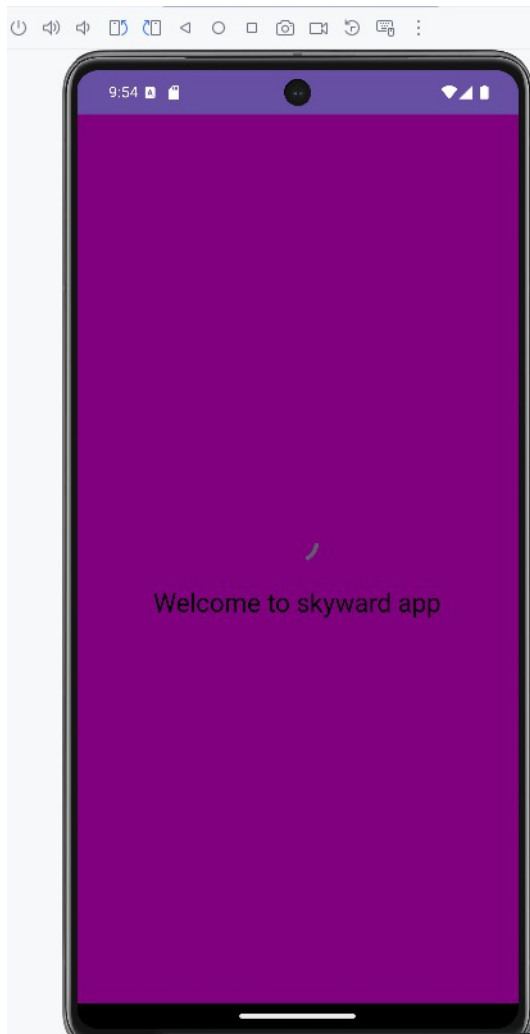
MainActivity.java

```

package com.example.splashscreen;
import android.os.Bundle;
import android.os.Handler;
import android.view.View;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
    private static final int SPLASH_DISPLAY_LENGTH=10000;

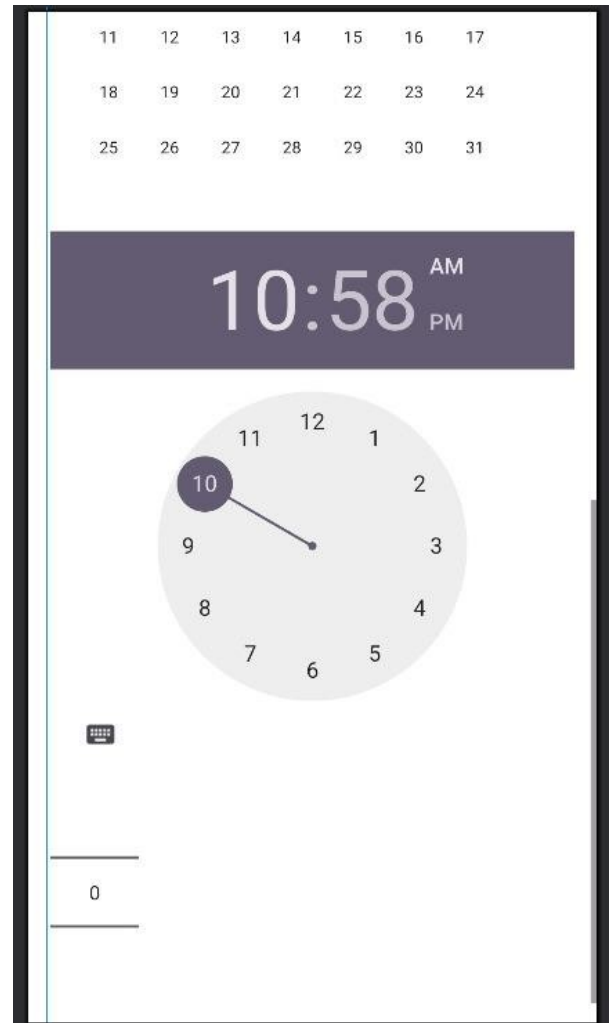
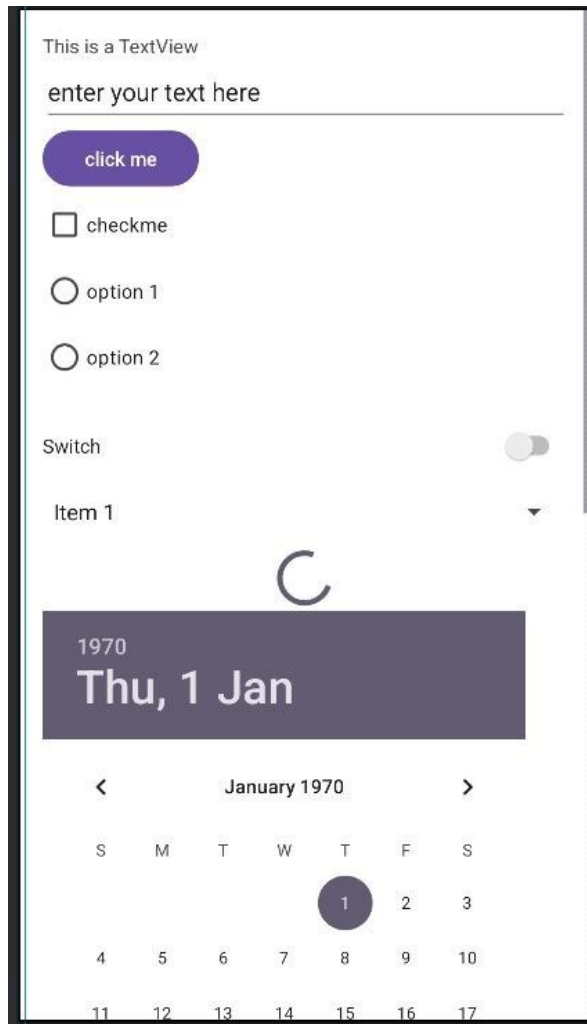
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        final View splashscreen=findViewById(R.id.splash_screen);
        final View mainContent=findViewById(R.id.main_content);
        new Handler().postDelayed(new Runnable()
        {
            @Override
            public void run()
            {
                splashscreen.setVisibility(View.GONE);
                mainContent.setVisibility(View.VISIBLE);
            }
        },SPLASH_DISPLAY_LENGTH);
    }
}

```

OUTPUT

Program 06: Create an UI with All Views.

Design



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="@color/white"
        android:orientation="vertical"
        android:padding="16dp">
```

```
<TextView
    android:id="@+id/textView"
    android:layout_width="243dp"
    android:layout_height="wrap_content"
    android:text="This is a TextView" />

<EditText
    android:id="@+id/editText"
    android:layout_width="381dp"
    android:layout_height="wrap_content"
    android:ems="10"
    android:inputType="text"
    android:minHeight="48dp"
    android:text="enter your text here" />

<Button
    android:id="@+id/button"
    android:layout_width="113dp"
    android:layout_height="wrap_content"
    android:text="click me" />

<CheckBox
    android:id="@+id/checkBox"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="checkme" />

<RadioGroup
    android:id="@+id/radioGroup"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:layout_marginBottom="16dp">

    <RadioButton
        android:id="@+id/radioButton"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="option 1" />

    <RadioButton
        android:id="@+id/radioButton3"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="option 2" />

</RadioGroup>
```

```
<Switch
    android:id="@+id/switch1"
    android:layout_width="378dp"
    android:layout_height="wrap_content"
    android:minHeight="48dp"
    android:text="Switch" />

<Spinner
    android:id="@+id/spinner"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:minHeight="48dp" />

<ProgressBar
    android:id="@+id/progressBar"
    style="?android:attr/progressBarStyle"
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />

<DatePicker
    android:id="@+id/datepicker"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>

<TimePicker
    android:id="@+id/timepicker"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"/>

<NumberPicker
    android:id="@+id/numberpicker"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginBottom="16dp"/>

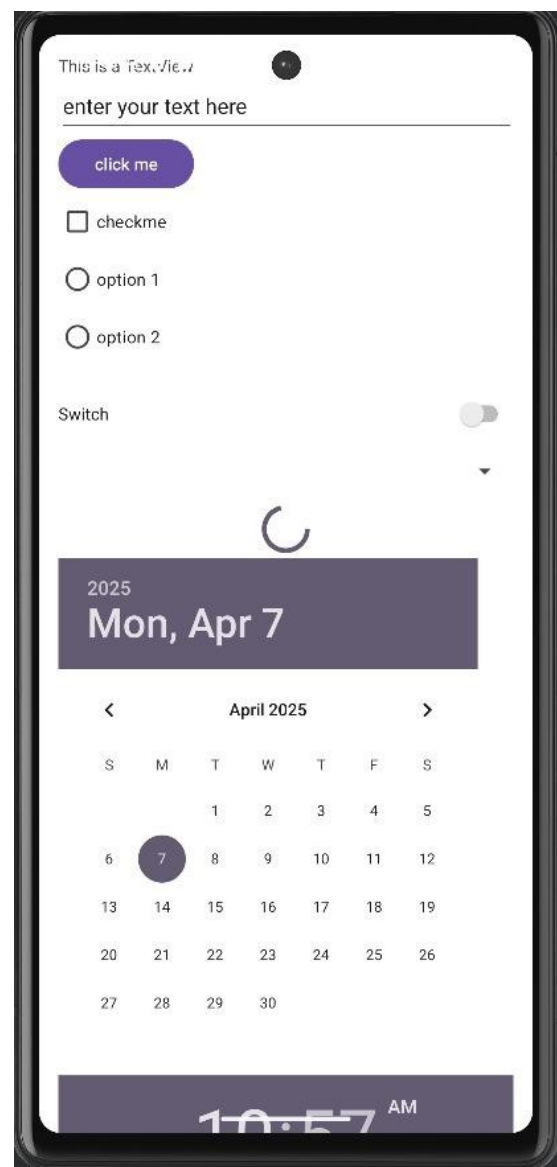
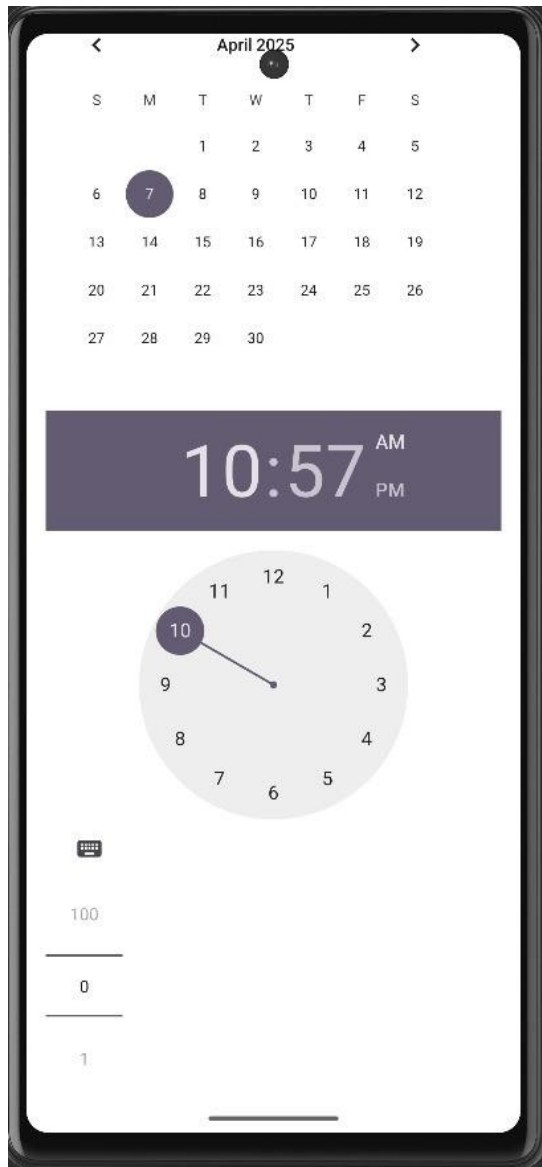
</LinearLayout>
</ScrollView>
```

MainActivity.java

```
package com.example.timedate;
import android.os.Bundle;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.DatePicker;
import android.widget.EditText;
import android.widget.NumberPicker;
import android.widget.ProgressBar;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.Spinner;
```

```
import android.widget.Switch;
import android.widget.TextView;
import android.widget.TimePicker;
import androidx.appcompat.app.AppCompatActivity;
import com.example.timedate.R;
public class MainActivity extends AppCompatActivity
{ private TextView textView;
  private EditText editText;
  private Button button;private CheckBox checkBox;
  private RadioGroup radioGroup;
  private RadioButton radioButton1;
  private RadioButton radioButton2;
  private Switch switch1;
  private Spinner spinner;
  private ProgressBar progressBar;
  private DatePicker datePicker;
  private TimePicker timePicker;
  private NumberPicker numberPicker;
  @Override
  protected void onCreate(Bundle savedInstanceState)
  { super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    textView = findViewById(R.id.textView);
    editText = findViewById(R.id.editText);
    button = findViewById(R.id.button);
    checkBox = findViewById(R.id.checkBox);
    radioGroup=findViewById(R.id.radioGroup);
    radioButton1 = findViewById(R.id.radioButton);
    radioButton2 =findViewById(R.id.radioButton3);
    switch1=findViewById(R.id.switch1);
    spinner =findViewById(R.id.spinner);
    progressBar = findViewById(R.id.progressBar);
    datePicker =findViewById(R.id.datepicker);
    timePicker = findViewById(R.id.timepicker);
    numberPicker =findViewById(R.id.numberpicker);
    numberPicker.setMinValue(0);
    numberPicker.setMaxValue (100);
  }
}
```

OUTPUT

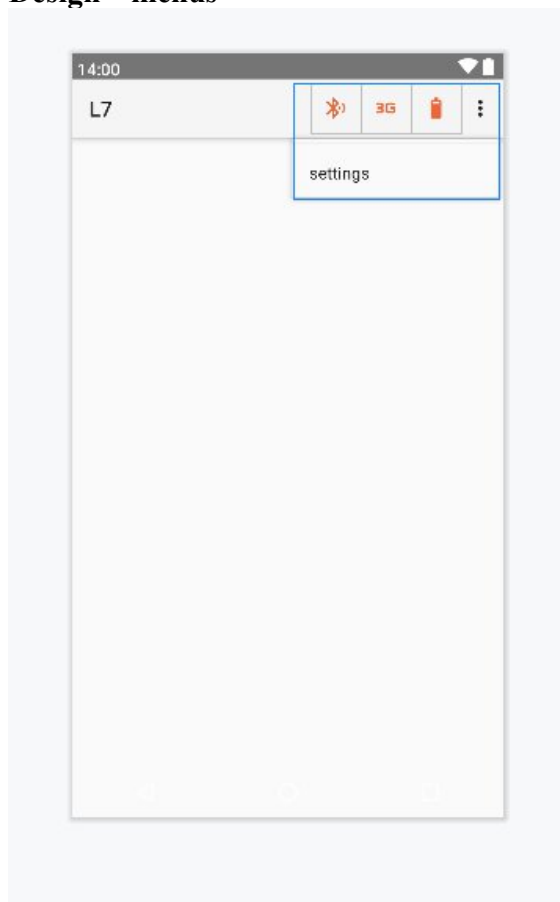


Program 07: Create menu in Application

Design - activity_main



Design – menus



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"

    android:background="@drawable/e_commerce"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="286dp"
        android:layout_height="65dp"
        android:layout_alignParentStart="true"
        android:layout_alignParentTop="true"
        android:layout_marginStart="59dp"
        android:layout_marginTop="63dp"
        android:text="Welcome to E-Commerce"
        android:textColor="#F11D1D"
        android:textColorLink="#F31818"
        android:textSize="24sp"
        android:textStyle="bold|italic" />

</RelativeLayout>
```

menus.xml

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto">
    <item
        android:id="@+id/search"
        android:icon="@drawable/search"
        app:showAsAction="always"
        android:title="search" />
    <item android:id="@+id/Refresh"
        android:title="Refresh"
        android:icon="@drawable/refresh"
        app:showAsAction="always" />
    <item android:id="@+id/Addtocart"
        android:title="Addtocart"/>
    <item android:id="@+id/settings"
        android:title="settings"/>
</menu>
```

MainActivity.java

```
package com.example.17;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuInflater;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
```

```

public class MainActivity extends AppCompatActivity
{
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
    @Override
    public boolean onCreatePanelMenu(int featureId, @NonNull Menu menu)
    {
        MenuInflater inflater = getMenuInflater();
        inflater.inflate(R.menu.menus, menu);
        return true;
    }
    @Override
    public boolean onOptionsItemSelected(@NonNull MenuItem item)
    {
        if (item.getItemId() == R.id.search) {
            Toast.makeText(this, "Search Successfull", Toast.LENGTH_SHORT).show();
        }
        if (item.getItemId() == R.id.Refresh) {
            Toast.makeText(this, "Refresh Successfull", Toast.LENGTH_SHORT).show();
        }
        if (item.getItemId() == R.id.Addtocart) {
            Toast.makeText(this, "Addtocart Successfull", Toast.LENGTH_SHORT).show();
        }
        if (item.getItemId() == R.id.settings) {
            Toast.makeText(this, "Settings Clicked", Toast.LENGTH_SHORT).show();
        }
        return super.onOptionsItemSelected(item);
    }
}

```

AndroidManifest.xml

```

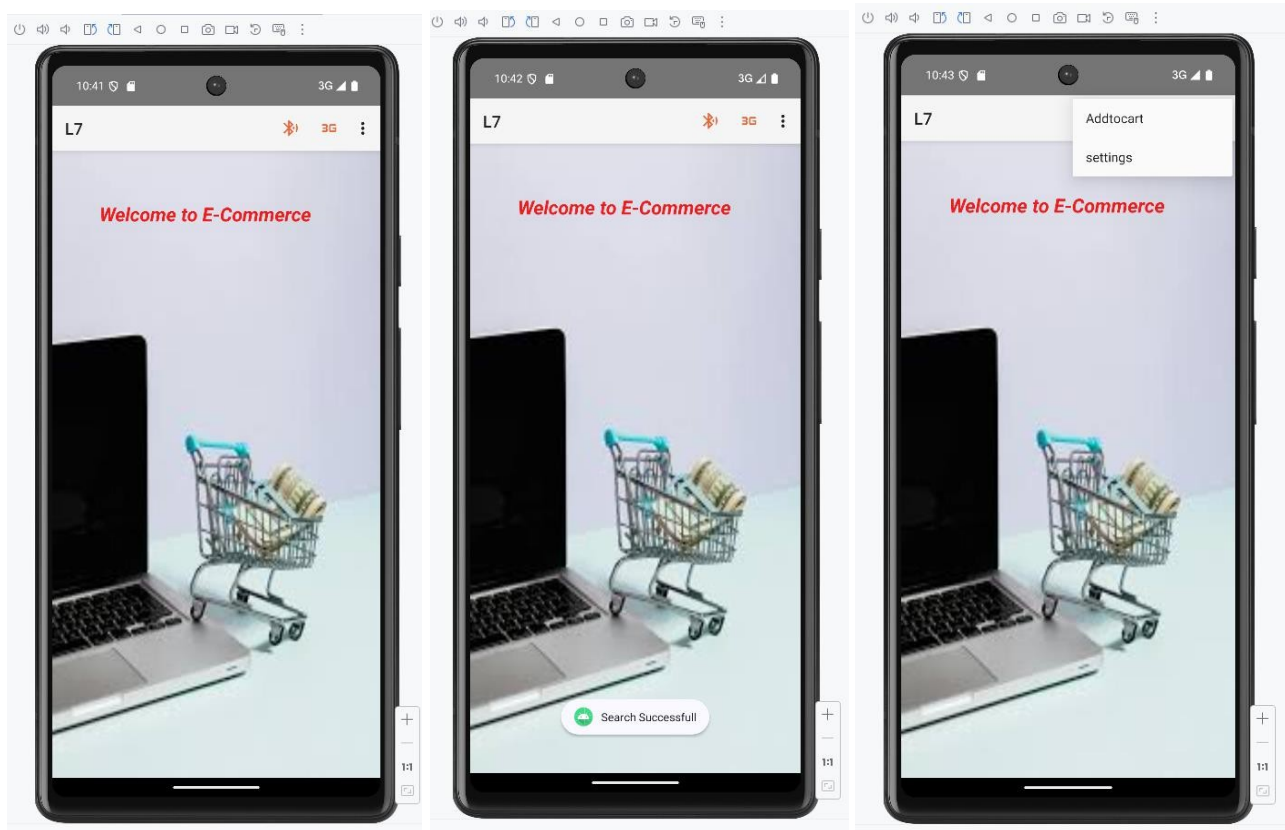
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/Theme.AppCompat.Light"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>

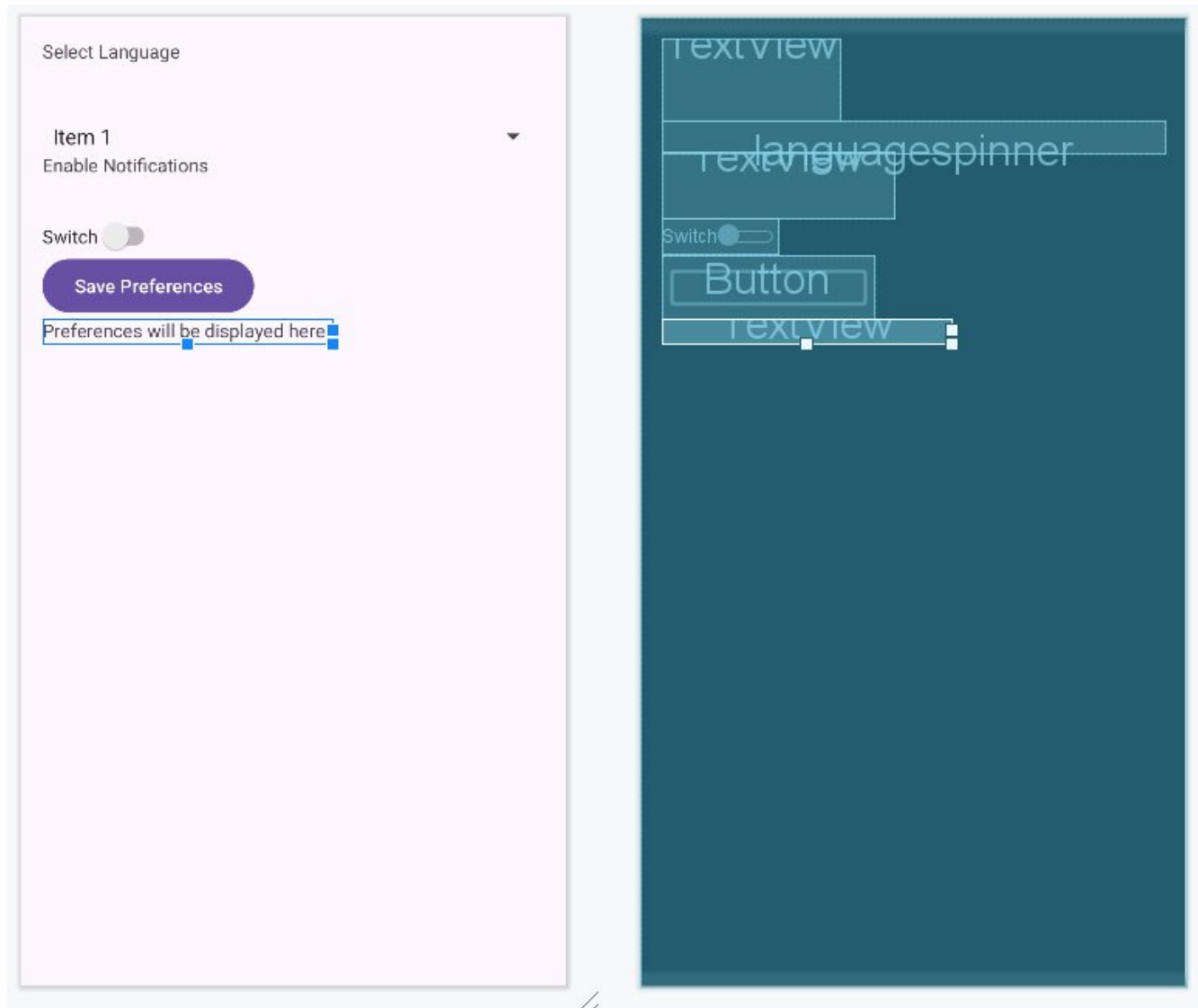
```

OUTPUT



Program 08: Read/Write the local data.

Design –



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:id="@+id/textView"
        android:layout_width="134dp"
        android:layout_height="62dp"
        android:layout_alignParentStart="true"
        android:layout_alignParentTop="true"
        android:text="Select Language" />
```

```

<Spinner
    android:id="@+id/languagespinner"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_alignParentTop="true"
    android:layout_alignParentEnd="true" />

<TextView
    android:id="@+id/textView2"
    android:layout_width="175dp"
    android:layout_height="50dp"
    android:layout_alignParentStart="true"
    android:layout_alignParentTop="true"
    android:text="Enable Notifications" />

<Switch
    android:id="@+id/notificationswitch"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentStart="true"
    android:layout_alignParentTop="true"
    android:text="Switch" />

<Button
    android:id="@+id/savebutton"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentStart="true"
    android:layout_alignParentTop="true"
    android:text="Save Preferences" />

<TextView
    android:id="@+id/preferenceTextView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_alignParentStart="true"
    android:layout_alignParentTop="true"
    android:text="Preferences will be displayed here." />
</LinearLayout>

```

String.xml

```

<resources>
    <string name="app_name">userprefrence</string>
    <string-array name="languages">
        <item>English</item>
        <item>Spanish</item>
        <item>French</item>
        <item>german</item>
        <item>Chinese</item>
    </string-array>
</resources>

```

MainActivity.java

```

package com.example.userpreference;
import android.content.Context;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.Spinner;
import android.widget.Switch;
import android.widget.TextView;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
public class MainActivity extends AppCompatActivity
{
    private Spinner languageSpinner;
    private Switch notificationsSwitch;
    private TextView preferencesTextView;
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        languageSpinner = findViewById(R.id.languagespinner);
        notificationsSwitch = findViewById(R.id.notificationswitch);
        preferencesTextView = findViewById(R.id.preferenceTextView);
        Button saveButton = findViewById(R.id.savebutton);
        loadUserPreferences();
        saveButton.setOnClickListener(v -> saveUserPreferences());
    }
    private void saveUserPreferences()
    {
        String language = languageSpinner.getSelectedItem().toString();
        boolean notificationsEnabled = notificationsSwitch.isChecked();
        SharedPreferences sharedPreferences = getSharedPreferences("UserPreferences",
Context.MODE_PRIVATE);
        SharedPreferences.Editor editor = sharedPreferences.edit();
        editor.putString("language", language);
        editor.putBoolean("notificationsEnabled", notificationsEnabled);
        editor.apply();
        preferencesTextView.setText(String.format("Language:%s\nNotificationsEnabled:%b", language,
notificationsEnabled));
    }
    private void loadUserPreferences()
    {
        SharedPreferences
sharedPreferences=getSharedPreferences("UserPreferences",Context.MODE_PRIVATE);
        String language=sharedPreferences.getString("language","English");
        boolean notificationsEnabled=sharedPreferences.getBoolean("notificationsEnabled",false);
        ArrayAdapter<CharSequence> adapter=ArrayAdapter.createFromResource(this,R.array.languages,
android.R.layout.simple_spinner_item);

```

```

adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);
languageSpinner.setAdapter(adapter);
int spinnerPosition=adapter.getPosition(language);
languageSpinner.setSelection(spinnerPosition);
notificationsSwitch.setChecked(notificationsEnabled);

```

```

preferencesTextView.setText(String.format("Language:%s\nNotificationsEnabled:%b",language,notificationsEnabled));

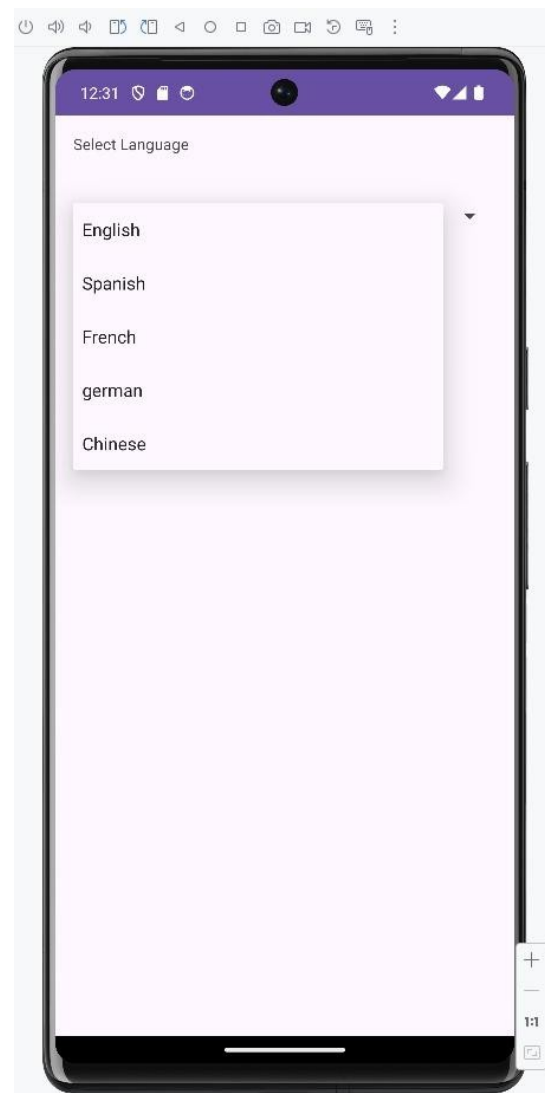
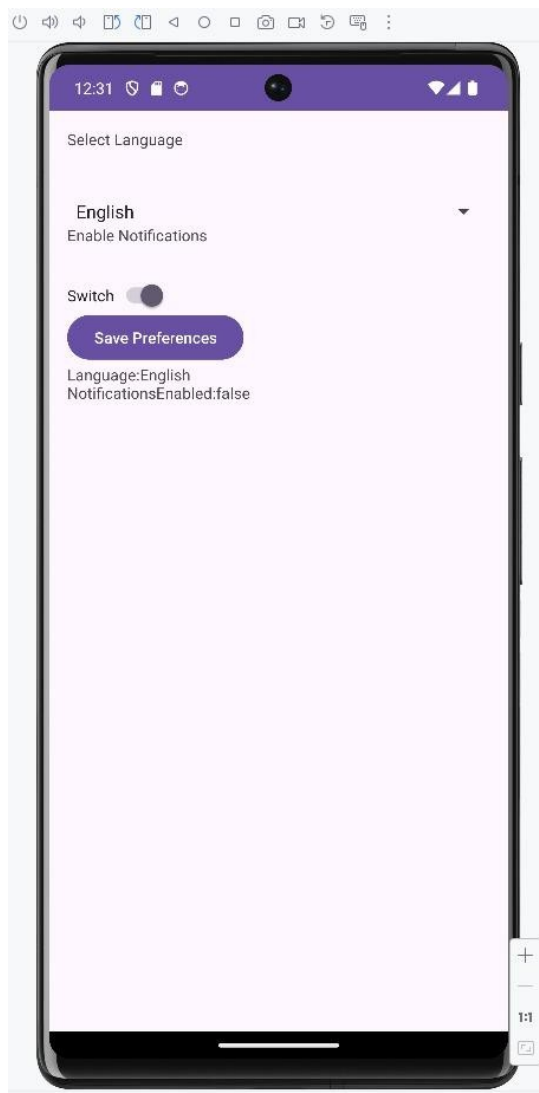
```

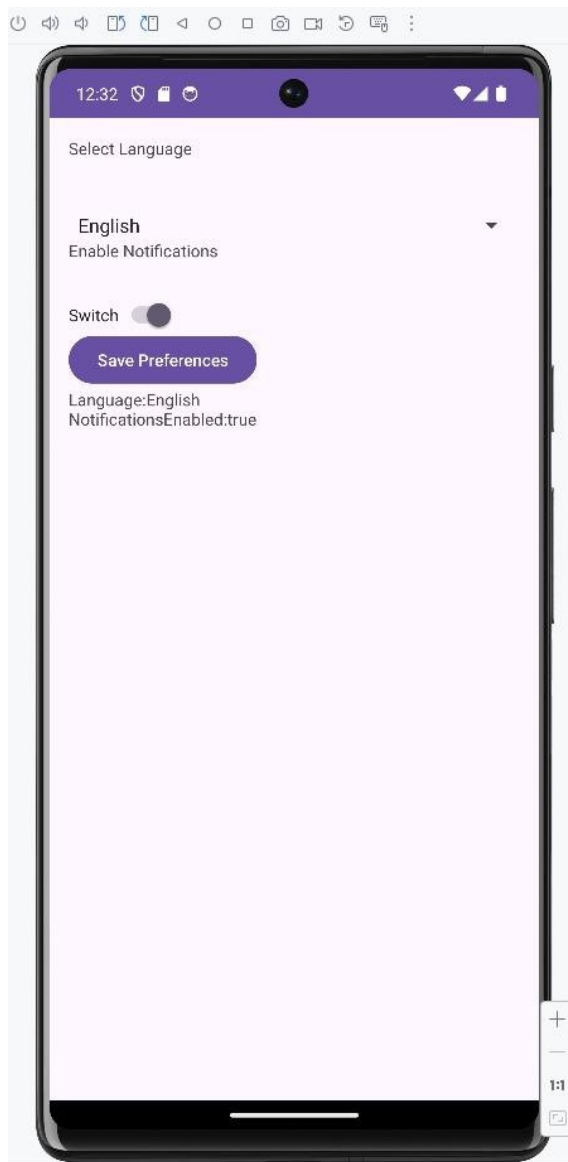
```

}
}

```

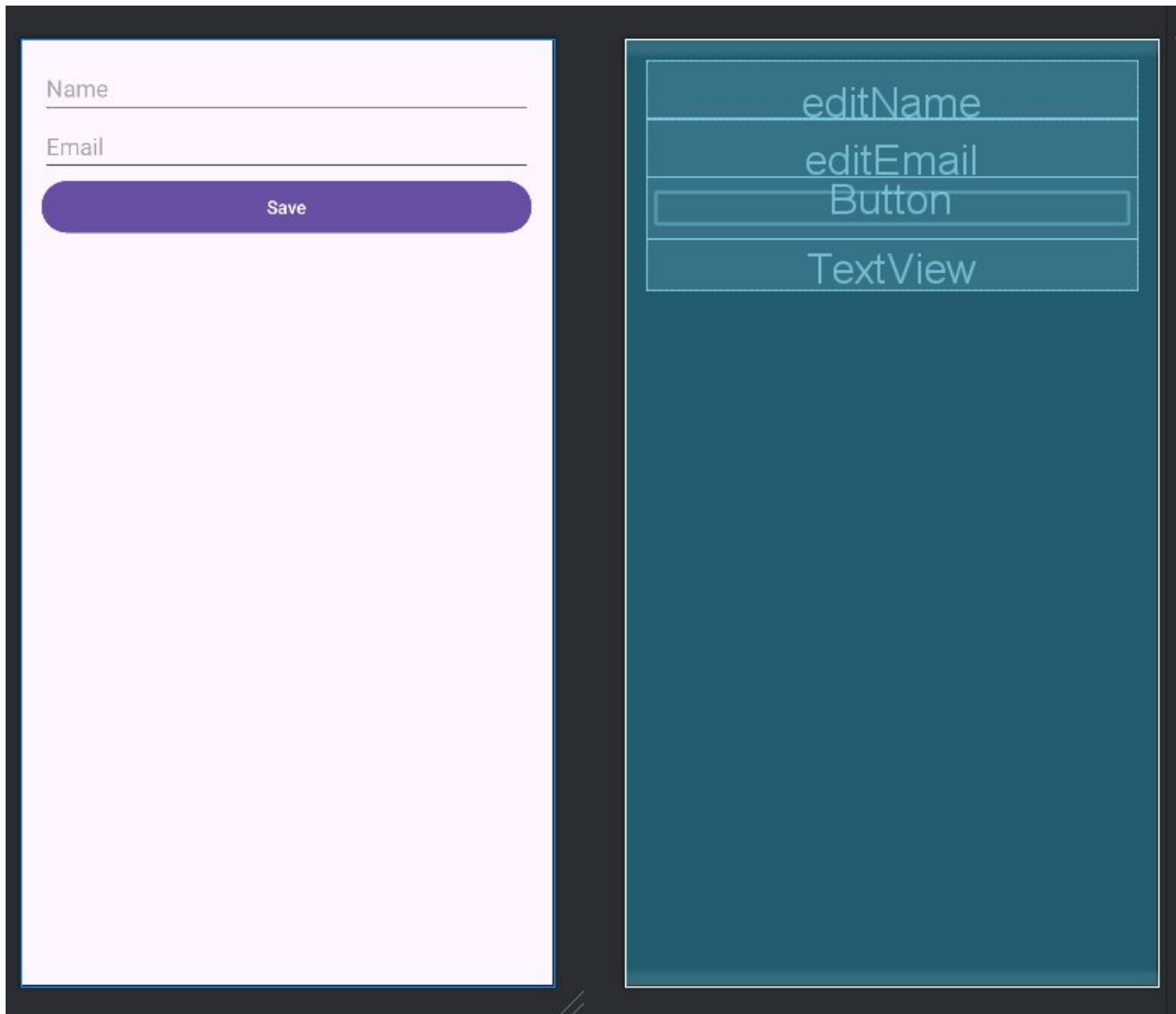
OUTPUT





Program 9: Create/Read/Write data with database(SQLite)

Design :



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="16dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
    <EditText
        android:id="@+id/editName"
        android:hint="Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <EditText
        android:id="@+id/editEmail"
        android:hint="Email"
        android:inputType="textEmailAddress"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>
    <Button
        android:id="@+id/buttonSave"
        android:text="Save"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />
    <TextView
```

```

        android:id="@+id/textViewResult"
        android:text=""
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:paddingTop="20dp"/>
</LinearLayout>

```

MainActivity.java

```

package com.example.db;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import android.widget.EditText;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
    EditText editName, editEmail;
    Button buttonSave;
    TextView textViewResult;
    SQLiteHelper dbHelper;
    @Override
    protected void onCreate(Bundle savedInstanceState)
    { super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      editName = findViewById(R.id.editName);
      editEmail = findViewById(R.id.editEmail);
      buttonSave = findViewById(R.id.buttonSave);
      textViewResult = findViewById(R.id.textViewResult);
      dbHelper = new SQLiteHelper(this);
      buttonSave.setOnClickListener(v -> {
          String name = editName.getText().toString();
          String email = editEmail.getText().toString();
          dbHelper.insertUser(name, email);
          String allUsers = dbHelper.getAllUsers();
          textViewResult.setText(allUsers);
      });
    }
}

```

SQLiteHelper.java

```

package com.example.db;
import android.database.sqlite.SQLiteDatabase;
import android.content.Context;
import android.content.ContentValues;
import android.database.sqlite.SQLiteOpenHelper;
import android.database.Cursor;
public class SQLiteHelper extends SQLiteOpenHelper
{ private static final String DB_NAME = "UserDB";
  private static final int DB_VERSION = 1;
  public SQLiteHelper(Context context)
  { super(context, DB_NAME, null,
    DB_VERSION);
  }
  @Override
  public void onCreate(SQLiteDatabase db) {
      db.execSQL("CREATE TABLE users (id INTEGER PRIMARY KEY AUTOINCREMENT, name
      TEXT, email TEXT)");
  }
  @Override

```

```

public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion)
{
    db.execSQL("DROP TABLE IF EXISTS users");
    onCreate(db);
}

public void insertUser(String name, String email)
{
    SQLiteDatabase db = this.getWritableDatabase();
    ContentValues values = new ContentValues();
    values.put("name", name);
    values.put("email", email);
    db.insert("users", null, values);
}

public String getAllUsers() {
    SQLiteDatabase db = this.getReadableDatabase();
    Cursor cursor = db.rawQuery("SELECT * FROM users", null);
    StringBuilder sb = new StringBuilder();

    while (cursor.moveToNext())
    {
        sb.append("ID: ")
        .append(cursor.getInt(0))
        .append(", Name: ")
        .append(cursor.getString(1))
        .append(", Email: ")
        .append(cursor.getString(2))
        .append("\n");
    }
    cursor.close();
    return sb.toString();
}
}

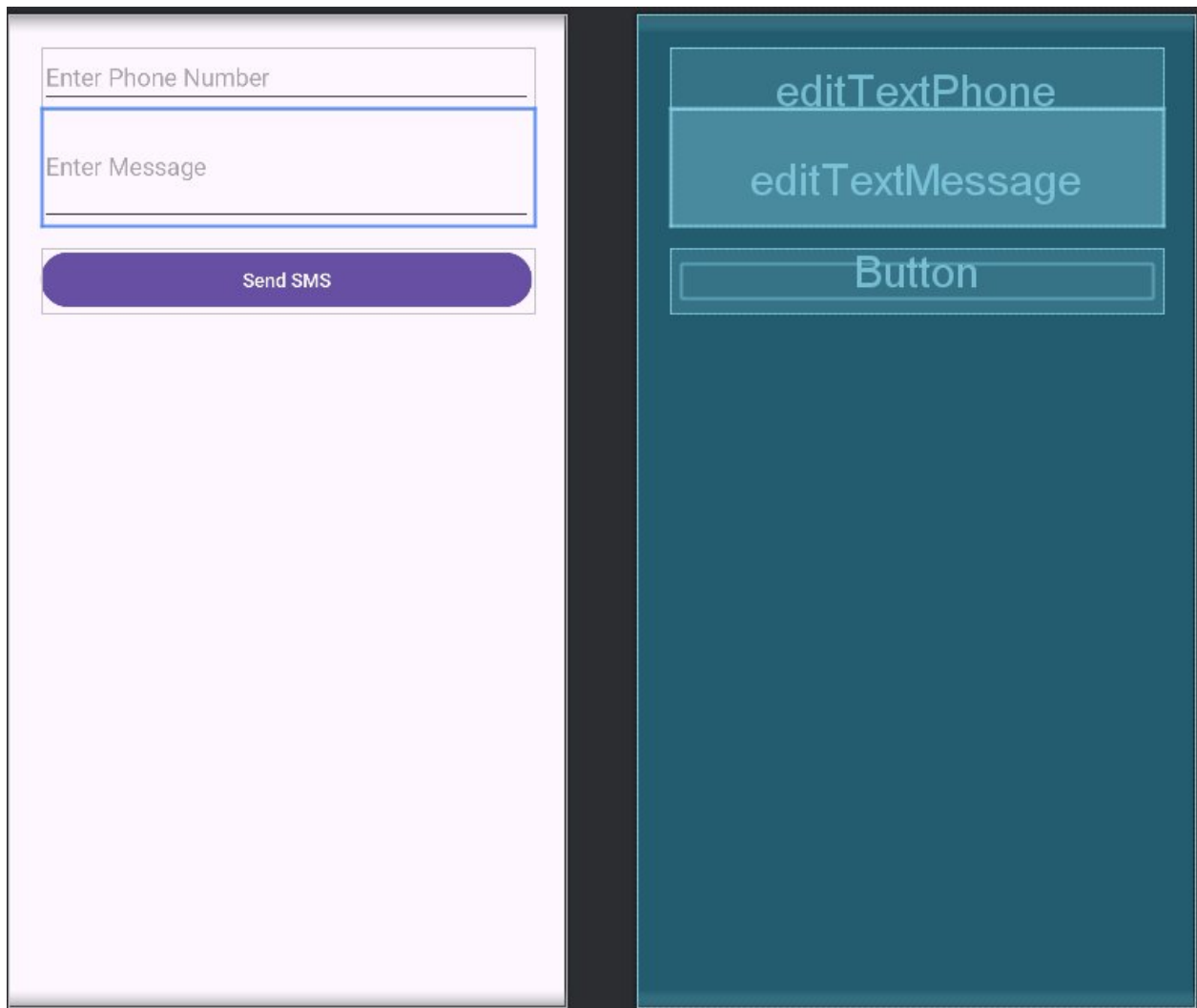
```

OUTPUT



Program 10: Create an application to send SMS and Receive SMS.

Design:



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="24dp">

    <EditText
        android:id="@+id/editTextPhone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Phone Number"
        android:inputType="phone"/>

    <EditText
        android:id="@+id/editTextMessage"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Message"
        android:inputType="textMultiLine"
    />

    <Button
        android:id="@+id/button"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Send SMS"
    />

</LinearLayout>
```

```
android:minLines="3"/>
```

```
<Button
    android:id="@+id/buttonSend"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Send SMS"
    android:layout_marginTop="16dp"/>
```

```
</LinearLayout>
```

MainActivity.java

```
package com.example.sms;
import android.Manifest;
import android.annotation.SuppressLint;
import android.content.pm.PackageManager;
import android.os.Bundle;
import android.telephony.SmsManager;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
public class MainActivity extends AppCompatActivity {

    EditText phoneNumberEditText, messageEditText;
    Button sendButton;
    private static final int SMS_PERMISSION_CODE = 100;
    private String permission;
    @SuppressWarnings("MissingInflatedId")
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        phoneNumberEditText = findViewById(R.id.editTextPhone);
        messageEditText = findViewById(R.id.editTextMessage);
        sendButton = findViewById(R.id.buttonSend);

        sendButton.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View v) {
                if (checkPermission(Manifest.permission.SEND_SMS))
                {
                    sendSMS();
                }
                else {
                    ActivityCompat.requestPermissions(MainActivity.this,
                        new String[]{Manifest.permission.SEND_SMS}, SMS_PERMISSION_CODE);
                }
            }
        });
    }

    private boolean checkPermission(String permission) {
```

```

        this.permission = permission;
        int check = ContextCompat.checkSelfPermission(this, permission);
        return (check == PackageManager.PERMISSION_GRANTED);
    }
    private void sendSMS() {
        String phoneNumber = phoneNumberEditText.getText().toString();
        String message = messageEditText.getText().toString();

        try {
            SmsManager smsManager = SmsManager.getDefault();
            smsManager.sendTextMessage(phoneNumber, null, message, null, null);
            Toast.makeText(this, "SMS Sent!", Toast.LENGTH_SHORT).show();
        } catch (Exception e) {
            Toast.makeText(this, "Failed to send SMS: " + e.getMessage(), Toast.LENGTH_LONG).show();
        }
    }
    @SuppressWarnings("MissingSuperCall")
    @Override

    public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions, @NonNull
int[] grantResults) {
        if (requestCode == SMS_PERMISSION_CODE) {
            if (grantResults.length > 0 && grantResults[0] == PackageManager.PERMISSION_GRANTED)
            {
                sendSMS();
            } else {
                Toast.makeText(this, "SMS permission denied", Toast.LENGTH_SHORT).show();
            }
        }
    }
}

```

AndroidManifest.xml

```

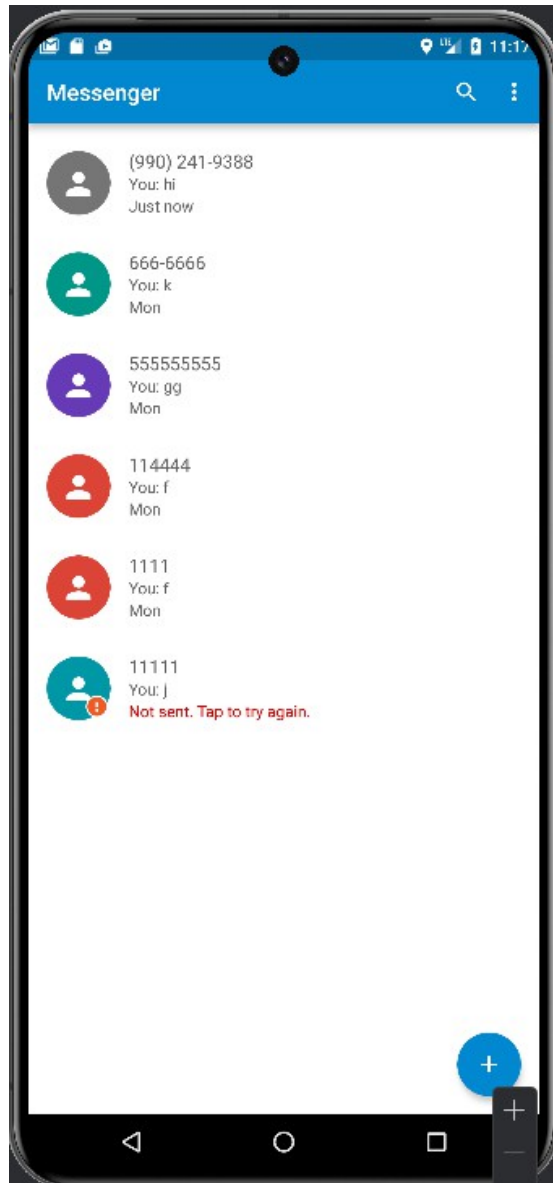
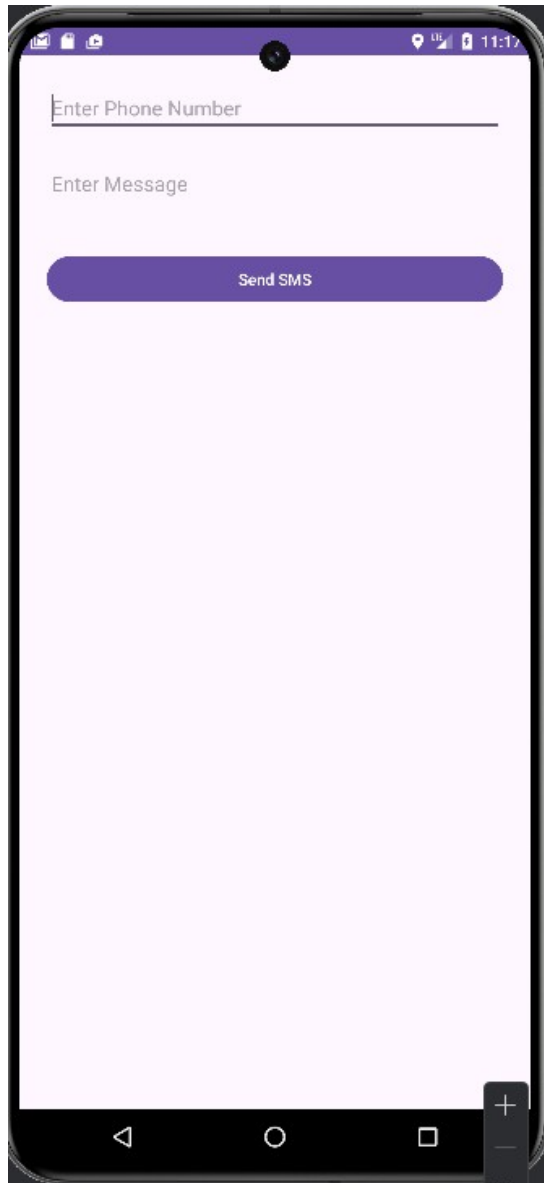
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.SEND_SMS"
        tools:ignore="PermissionImpliesUnsupportedChromeOsHardware" />
    <uses-permission android:name="android.permission.READ_PHONE_STATE"/>
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/Theme.Sms"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>

```

```
<action android:name="android.intent.action.MAIN" />

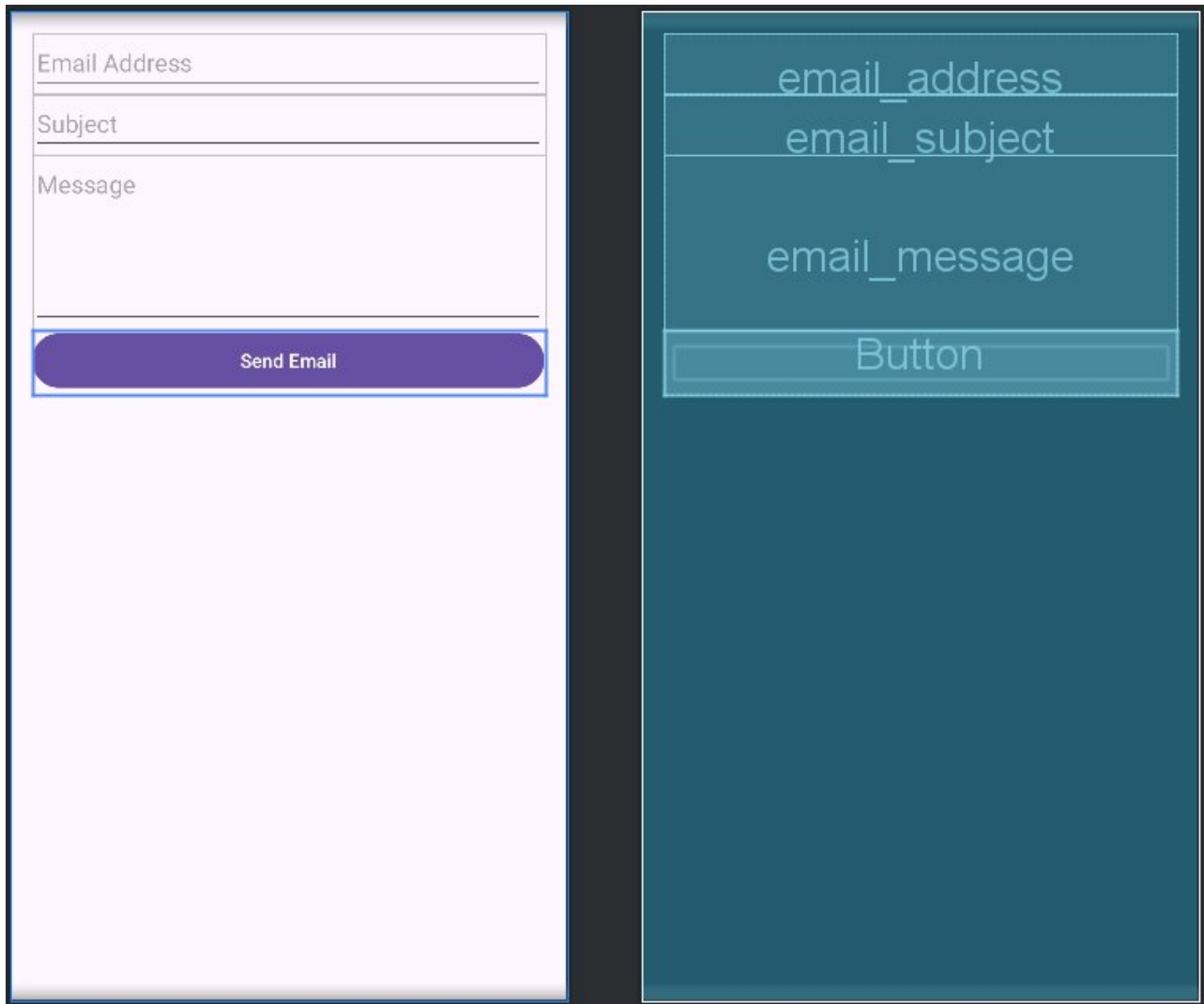
<category android:name="android.intent.category.LAUNCHER" />
</intent-filter>
</activity>
</application>
</manifest>
```

OUTPUT



Program 11: Create an application to send a Email.

Design



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/email_address"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Email Address"
        android:inputType="textEmailAddress" />

    <EditText
        android:id="@+id/email_subject"
```



```

        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Subject"
        android:inputType="text" />

```

```
<EditText
```

```

        android:id="@+id/email_message"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Message"
        android:inputType="textMultiLine"
        android:lines="5"
        android:gravity="top"
        android:scrollbars="vertical" />

```

```
<Button
```

```

        android:id="@+id/send_email"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Send Email" />

```

```
</LinearLayout>
```

MainActivity.java

```

package com.example.email;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import com.example.email.R;

public class MainActivity extends AppCompatActivity {

    private EditText emailAddressEditText;
    private EditText emailSubjectEditText;
    private EditText emailMessageEditText;
    private Button sendEmailButton;

    @Override
    protected void onCreate(Bundle savedInstanceState) {

```

```

super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
emailAddressEditText = findViewById(R.id.email_address);
emailSubjectEditText = findViewById(R.id.email_subject);
emailMessageEditText = findViewById(R.id.email_message);
sendEmailButton = findViewById(R.id.send_email);
sendEmailButton.setOnClickListener(new View.OnClickListener()
    { @Override
      public void onClick(View v)
      { sendEmail();
      }
    });
}

private void sendEmail() {
    String emailAddress = emailAddressEditText.getText().toString();
    String subject = emailSubjectEditText.getText().toString();
    String message = emailMessageEditText.getText().toString();
    Intent emailIntent = new Intent(Intent.ACTION_SEND);
    emailIntent.setType("message/rfc822");
    emailIntent.putExtra(Intent.EXTRA_EMAIL, new String[] {emailAddress});
    emailIntent.putExtra(Intent.EXTRA_SUBJECT, subject);
    emailIntent.putExtra(Intent.EXTRA_TEXT, message);

    if (emailIntent.resolveActivity(getPackageManager()) != null)
        { startActivity(Intent.createChooser(emailIntent, "Send Email Using:"));
        } else {
        Toast.makeText(this, "Email Client Not Installed", Toast.LENGTH_SHORT).show();
        }
    }
}

```

AndroidManifest.xml

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.SEND_SMS"
        tools:ignore="PermissionImpliesUnsupportedChromeOsHardware" />
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"

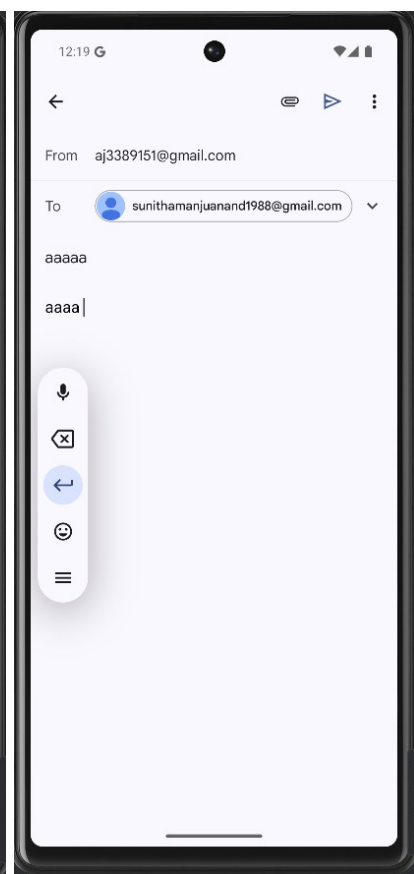
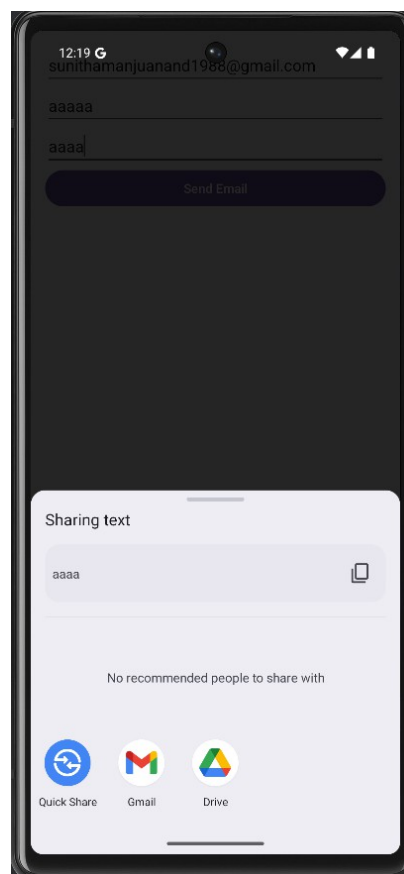
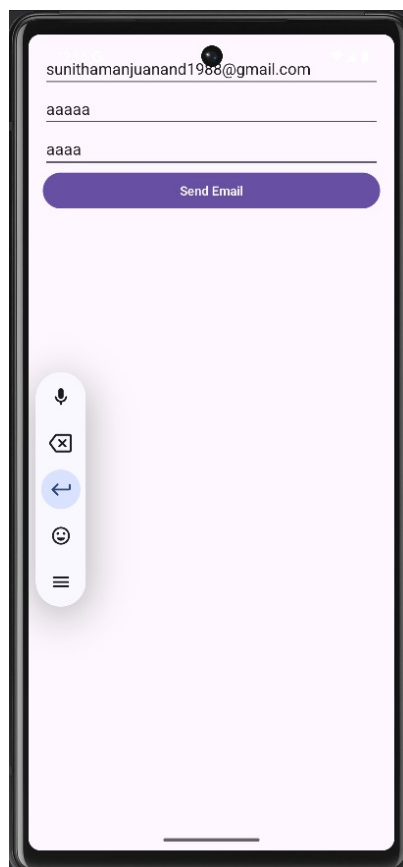
```

```

    android:label="@string/app_name"
    android:roundIcon="@mipmap/ic_launcher_round"
    android:supportsRtl="true"
    android:theme="@style/Theme.Email"
    tools:targetApi="31">
    <activity
        android:name=".MainActivity"
        android:exported="true">
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>
</application>
</manifest>

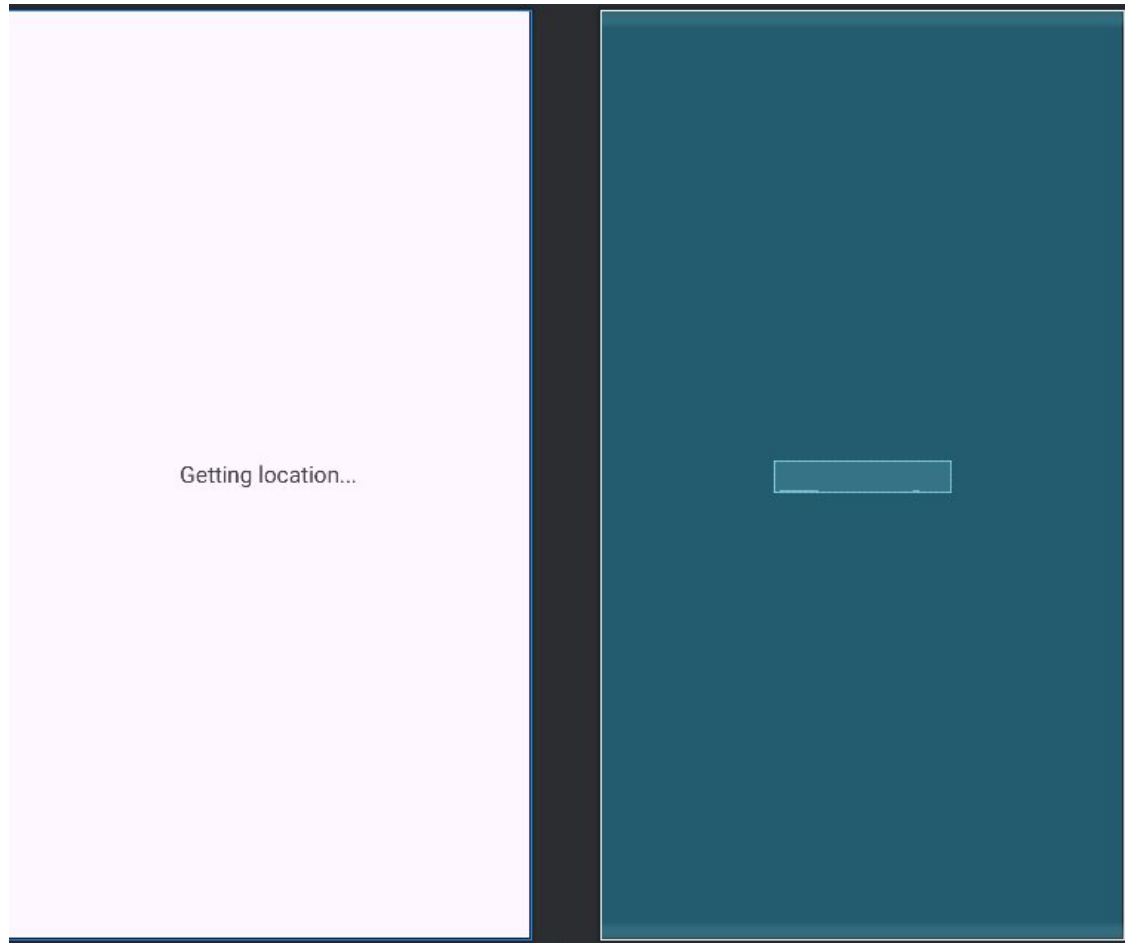
```

OUTPUT



Program 12: Display Map based on the Current/given location.

Design



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <TextView
        android:id="@+id/location_text"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Getting location..."
        android:textSize="18sp" />
</LinearLayout>
```

MainActivity.java

```
package com.example.loc;
import android.Manifest;
import android.content.pm.PackageManager;
import android.location.Location;
import android.location.LocationListener;
import android.location.LocationManager;
import android.os.Bundle;
```

```

import android.widget.TextView;
import android.widget.Toast;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;

public class MainActivity extends AppCompatActivity {
    private static final int LOCATION_PERMISSION_CODE = 1;
    private LocationManager locationManager;
    private LocationListener locationListener;
    private TextView locationText;
    @Override
    protected void onCreate(Bundle savedInstanceState)
    {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        locationText = findViewById(R.id.location_text);
        locationManager = (LocationManager) getSystemService(LOCATION_SERVICE);
        locationListener = new LocationListener() {
            @Override
            public void onLocationChanged(@NonNull Location location)
            {
                double lat = location.getLatitude();
                double lon = location.getLongitude();
                locationText.setText("Latitude: " + lat + "\nLongitude: " + lon);
            }
        };
        // Check permissions
        if (ActivityCompat.checkSelfPermission(this, Manifest.permission.ACCESS_FINE_LOCATION)
!= PackageManager.PERMISSION_GRANTED)
        {
            ActivityCompat.requestPermissions(this,
                new String[]{Manifest.permission.ACCESS_FINE_LOCATION},
LOCATION_PERMISSION_CODE);
        } else {
            startLocationUpdates();
        }
    }
    private void startLocationUpdates() {
        if (ActivityCompat.checkSelfPermission(this, Manifest.permission.ACCESS_FINE_LOCATION)
== PackageManager.PERMISSION_GRANTED)
        {
            locationManager.requestLocationUpdates(
                LocationManager.GPS_PROVIDER,
                5000, // Time interval (ms)
                5, // Distance (meters)
                locationListener);
        }
    }
    @Override
    public void onRequestPermissionsResult(int requestCode, @NonNull String[] permissions, @NonNull
int[] grantResults) {
        super.onRequestPermissionsResult(requestCode, permissions, grantResults);

        if (requestCode == LOCATION_PERMISSION_CODE) {
            if (grantResults.length > 0 && grantResults[0] == PackageManager.PERMISSION_GRANTED)
        {
            startLocationUpdates();

```

```

    } else {
        Toast.makeText(this, "Location permission denied.", Toast.LENGTH_SHORT).show();
    }
}
}
}

```

AndroidManifest.xml

```

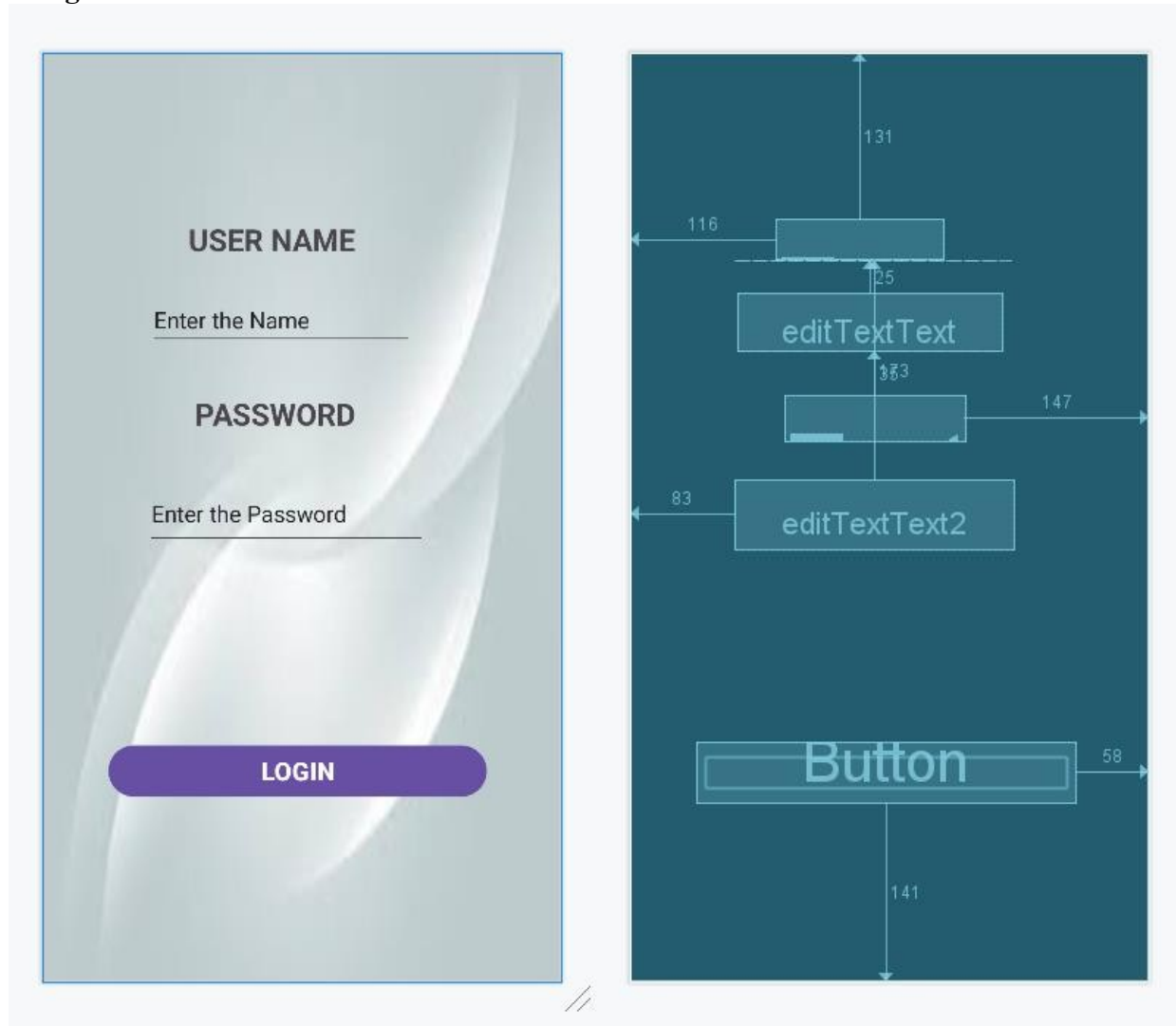
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />
    <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
    <uses-permission android:name="android.permission.INTERNET"/>
    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/Theme.Loc"
        tools:targetApi="31">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
</manifest>

```

OUTPUT

Program 13: Create a sample application with login module (check user name and password) On successful login change Textview “Login Successful”. On login fail alert using Toast “login fail”

Design



activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/login1"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_alignParentStart="true"
        android:layout_alignParentTop="true"
        android:layout_marginStart="116dp"
        android:layout_marginTop="131dp"
        android:text="USER NAME"
```



```
        android:textSize="24sp"
        android:textStyle="bold" />

<EditText
    android:id="@+id/editTextText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/textView"
    android:layout_marginStart="85dp"
    android:layout_marginTop="25dp"
    android:ems="10"
    android:hint="username"
    android:inputType="text"
    android:text="Enter the Name" />

<TextView
    android:id="@+id/textView2"
    android:layout_width="143dp"
    android:layout_height="36dp"
    android:layout_below="@id/editTextText"
    android:layout_alignParentEnd="true"
    android:layout_marginTop="35dp"
    android:layout_marginEnd="147dp"
    android:text="PASSWORD"
    android:textSize="24sp"
    android:textStyle="bold" />

<EditText
    android:id="@+id/editTextText2"
    android:layout_width="222dp"
    android:layout_height="55dp"
    android:layout_below="@id/textView"
    android:layout_alignParentStart="true"
    android:layout_marginStart="83dp"
    android:layout_marginTop="173dp"
    android:ems="10"
    android:hint="password"
    android:inputType="text"
    android:text="Enter the Password" />

<Button
    android:id="@+id/button"
    android:layout_width="301dp"
    android:layout_height="wrap_content"
    android:layout_alignParentEnd="true"
    android:layout_alignParentBottom="true"
    android:layout_marginEnd="58dp"
    android:layout_marginBottom="141dp"
    android:text="LOGIN"
    android:textSize="20sp"
    android:textStyle="bold" />
</RelativeLayout>
```

MainActivity.java

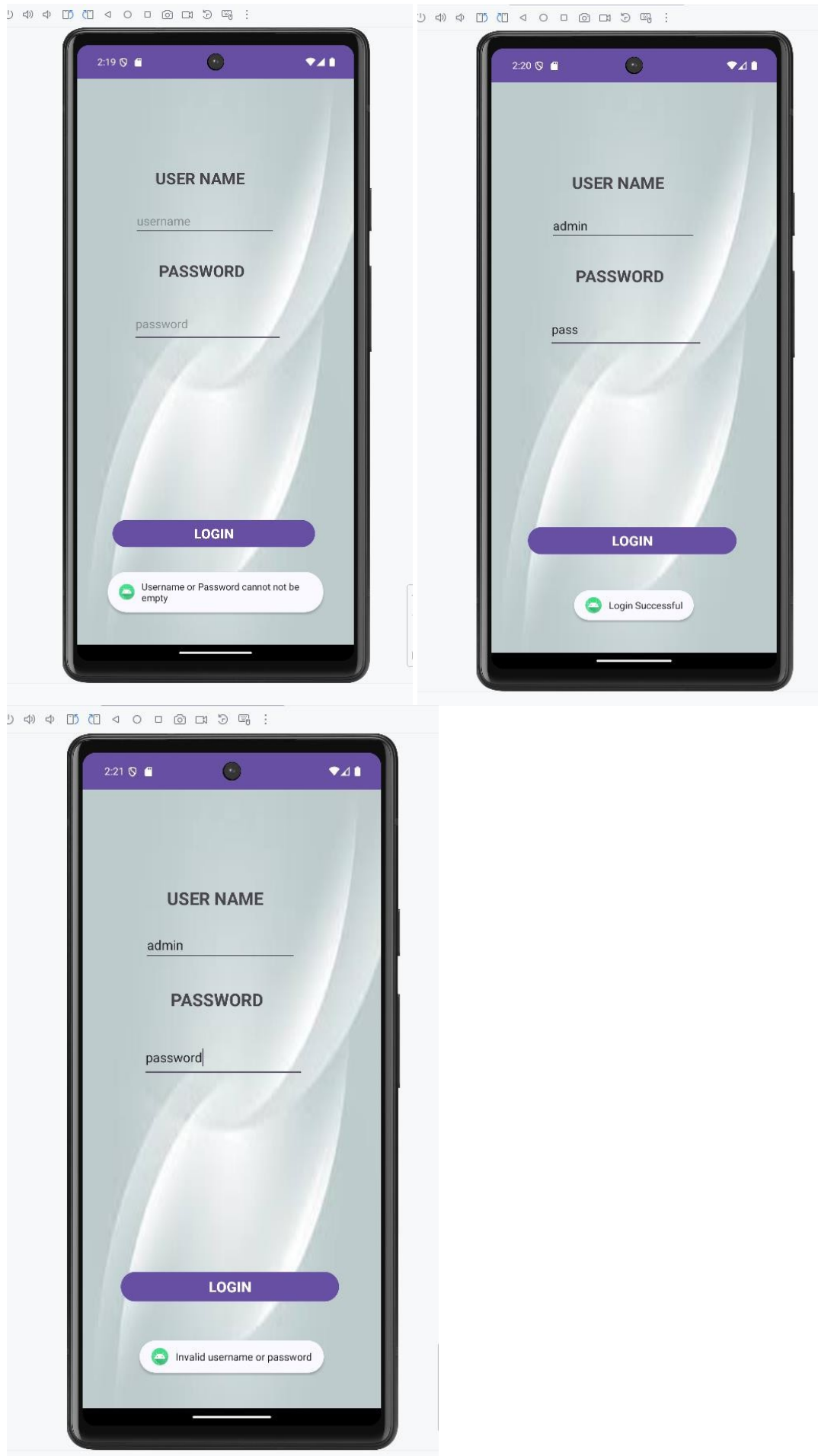
```
package com.example.loginapp;
import android.os.Bundle;
import android.text.TextUtils;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
```

```
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    private EditText editTextUsername, editTextPassword; private Button buttonlogin;
    @Override
    protected void onCreate(Bundle savedInstanceState) { super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        editTextUsername=findViewById(R.id.editTextText);
        editTextPassword=findViewById(R.id.editTextText2);
        buttonlogin=findViewById(R.id.button);

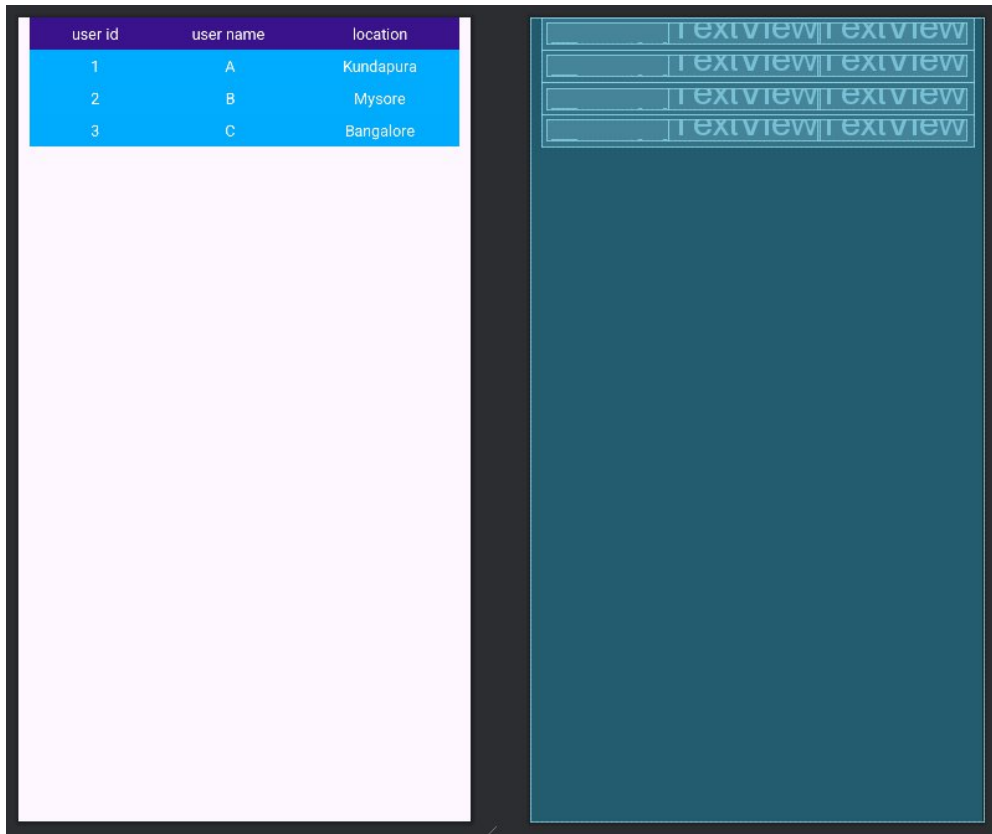
        buttonlogin.setOnClickListener(new View.OnClickListener()
        { @Override
        public void onClick(View v)
        {
            String username=editTextUsername.getText().toString().trim();
            String password=editTextPassword.getText().toString().trim();
            if(TextUtils.isEmpty(username)||TextUtils.isEmpty(password))
            {
                Toast.makeText(MainActivity.this," Username or Password
cannot not be empty",Toast.LENGTH_SHORT).show();
            }
            else
            {
                if (username.equals("admin")&& password.equals("pass"))
                {
                    Toast.makeText(MainActivity.this, "Login Successful", Toast.LENGTH_SHORT).show();
                } else
                {
                    Toast.makeText(MainActivity.this,"Invalid username or password",
Toast.LENGTH_SHORT).show();
                }
            }
        }
    });
}
```

OUTPUT



Program 14: Create an application to demonstrate table layout.

Design:



Code:

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<TableLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:paddingLeft="10dp"
    android:paddingRight="10dp"
    tools:context=".MainActivity">

    <TableRow android:background="#3A128C" android:padding="5dp">
        <TextView android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_weight="1" android:text="user id"
            android:textColor="@color/white"
            android:textAlignment="center"/>

        <TextView android:layout_width="wrap_content"
            android:layout_height="wrap_content"
```

```
        android:layout_weight="1" android:text="user
        name " android:textColor="@color/white"
        android:textAlignment="center"/>

        <TextView android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_weight="1" android:text="location"
        android:textColor="@color/white"
        android:textAlignment="center"/>
    </TableRow>

    <TableRow android:background="#00ACFF" android:padding="5dp">
        <TextView android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_weight="1" android:text="1"
        android:textColor="@color/white"
        android:textAlignment="center"/>

        <TextView android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_weight="1" android:text="A"
        android:textColor="@color/white"
        android:textAlignment="center"/>

        <TextView android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_weight="1" android:text="Kundapura"
        android:textColor="@color/white"
        android:textAlignment="center"/>
    </TableRow>

    <TableRow android:background="#01ACFF" android:padding="5dp">
        <TextView android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="2"
        android:textColor="@color/white"
        android:textAlignment="center"/>

        <TextView android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_weight="1" android:text="B"
        android:textColor="@color/white"
        android:textAlignment="center"/>

        <TextView android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_weight="1" android:text="Mysore"
        android:textColor="@color/white"
        android:textAlignment="center"/>
    </TableRow>

    <TableRow android:background="#00AAFD" android:padding="5dp">
```

```
<TextView android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_weight="1" android:text="3"
    android:textColor="@color/white"
    android:textAlignment="center"/>

<TextView android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_weight="1" android:text="C"
    android:textColor="@color/white"
    android:textAlignment="center"/>

<TextView android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_weight="1" android:text="Bangalore"
    android:textColor="@color/white"
    android:textAlignment="center"/>
```

```
</TableRow>
```

```
</TableLayout>
```

MainActivity.java

```
package com.example.demo11;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity; import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
```

```
public class MainActivity extends AppCompatActivity
{ @Override
protected void onCreate(Bundle savedInstanceState)
{
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
});
}
```

Output:A smartphone screen displaying a table with user data. The table has three columns: 'user id', 'user name', and 'location'. The data rows are highlighted in blue. The background of the screen is light pink.

user id	user name	location
1	A	Kundapura
2	B	Mysore
3	C	Bangalore